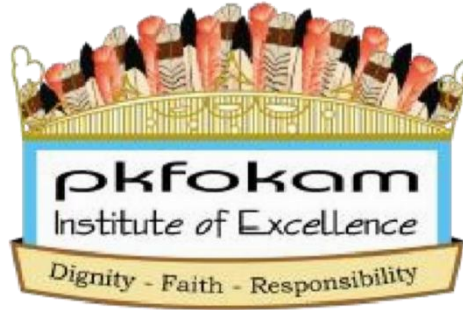




PKFOKAM INSTITUTE OF EXCELLENCE

PROJECT: WEDDING INVITATION AND GIFT MANAGEMENT SYSTEM

PROJECT REPORT

GIT HUB REPOSITORY AND PERSON MANAGEMENT CREATION

Course: C Programming

Group Members	Participation	Matricules
Matchim Dzuyim Verenne Celia (Group Leader)	22%	26S031
Djomo Yepdi Evans	18%	26S001
Talla Tchossi Daniella Elsa	22%	26S018
Mekolle Tuatsop Dalhia Aure	20%	26S004
Messi Zang Paul Jacques	18%	26S005

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Lecturer: Mr. Mbietieu Amos Mbietieu

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1. Project Overview

The **Wedding Invitation and Gift Management System (WIGMS)** is a software application designed to assist in organizing and managing wedding events. Weddings in Cameroon often involve a large number of guests, gift contributions, and multiple social categories, which makes manual management difficult.

The purpose of this system is to simplify wedding planning by providing tools to manage:

- Guest information
- Social categories of guests
- Invitation priority
- Wedding gift tracking

The application is developed using the **C programming language** and follows a **modular programming approach** where each part of the system is implemented as an independent module.

The system aims to improve organization, reduce errors, and help wedding planners efficiently manage guest lists and gifts.

Tool	Purpose
C Programming Language	Main programming language used for system implementation
GCC Compiler	Used to compile and run the C programs
Visual Studio Code / CodeBlocks	Code editor for writing and debugging the program
GitHub	Version control and collaboration platform
Standard C Libraries (stdio.h, stdlib.h,	Used for input/output operations, memory management and string handling

string.h)	
-----------	--

Table 1: Software tools and technologies used

2. Development Tools Used

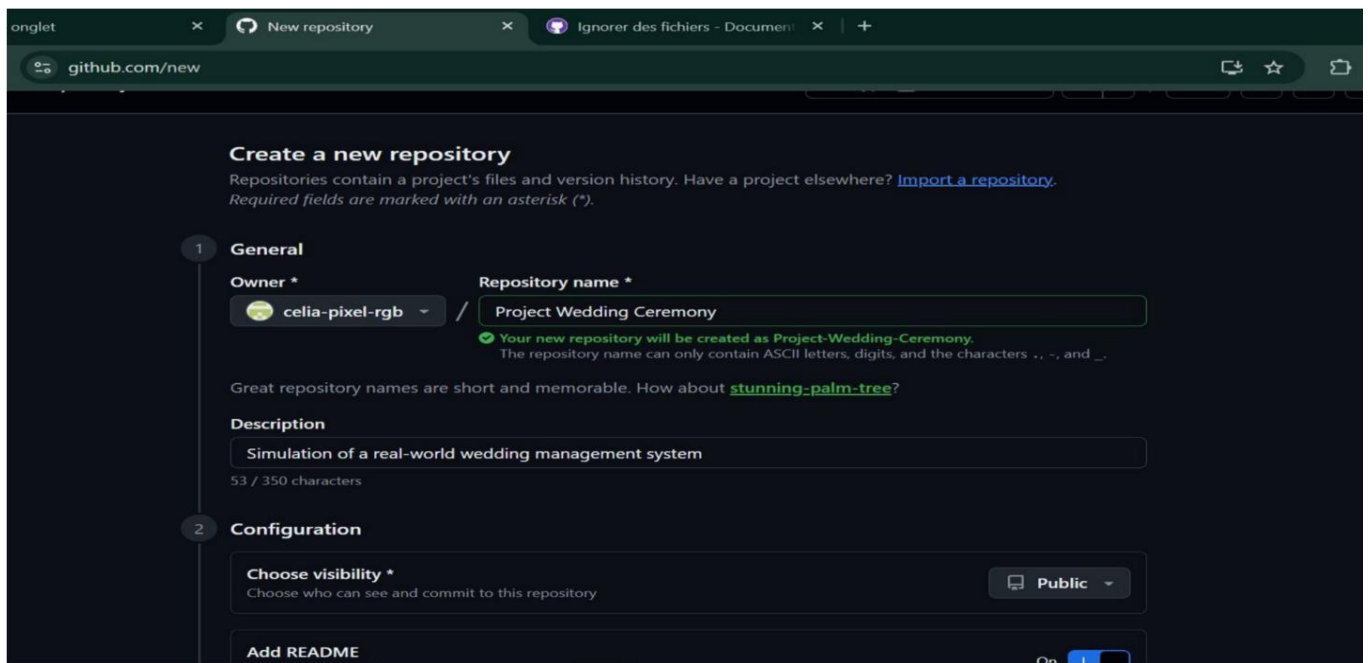
The following tools were used during the development of the system:

Table 2: GitHub repository configuration settings

SETTINGS	VALUE CHOSEN
Repository name	Project-Wedding-Ceremony
Visibility	Public
.gitignore	C template selected
Repository link	https://github.com/celia-pixelrgb/Project-Wedding-Ceremony

3. Github repository

3.1 Creating GitHub repository



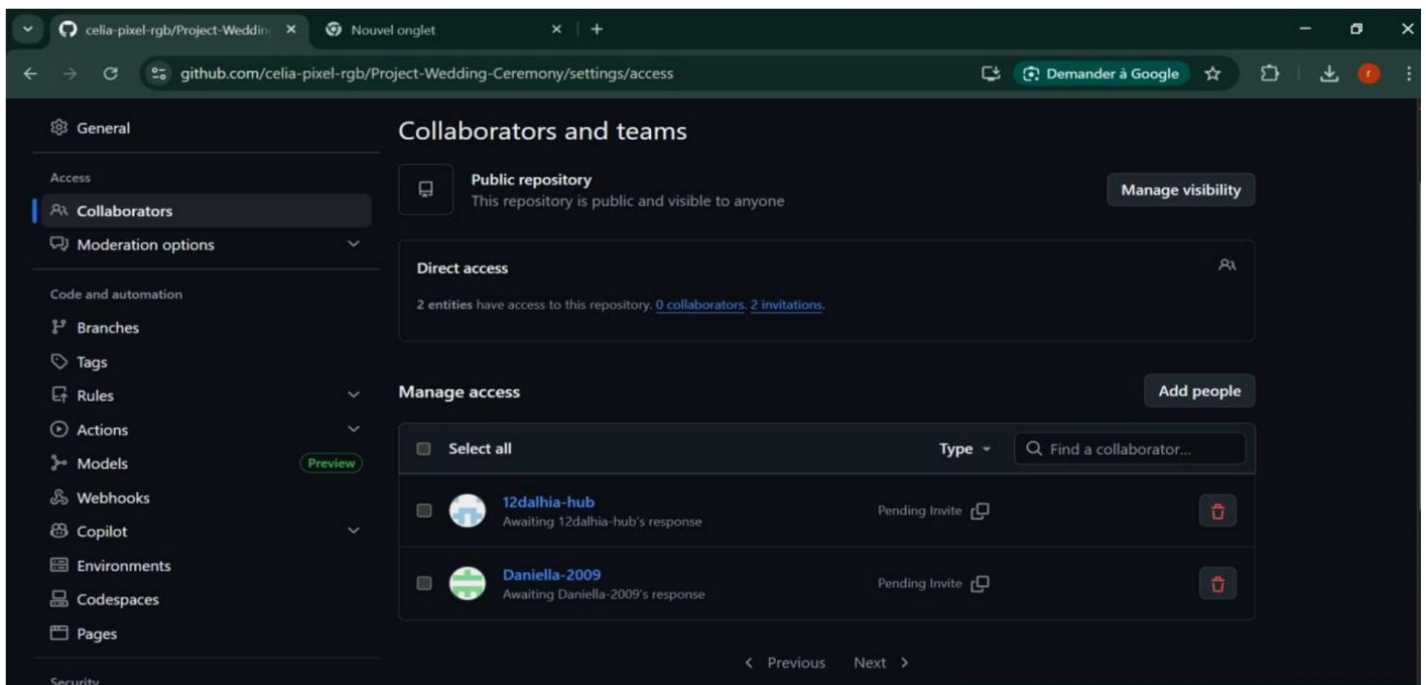
The group leader created the public repository with the following settings:

Here is the GitHub link to access our repository:

<https://github.com/celia-pixel-rgb/Project-Wedding-Ceremony>

3.2. Adding Collaborators

The remaining group members were added as collaborators through the Setting --» collaborators page of the repository. Each member received an Email invitation from GitHub which they accepted in order to gain push access to the repository.



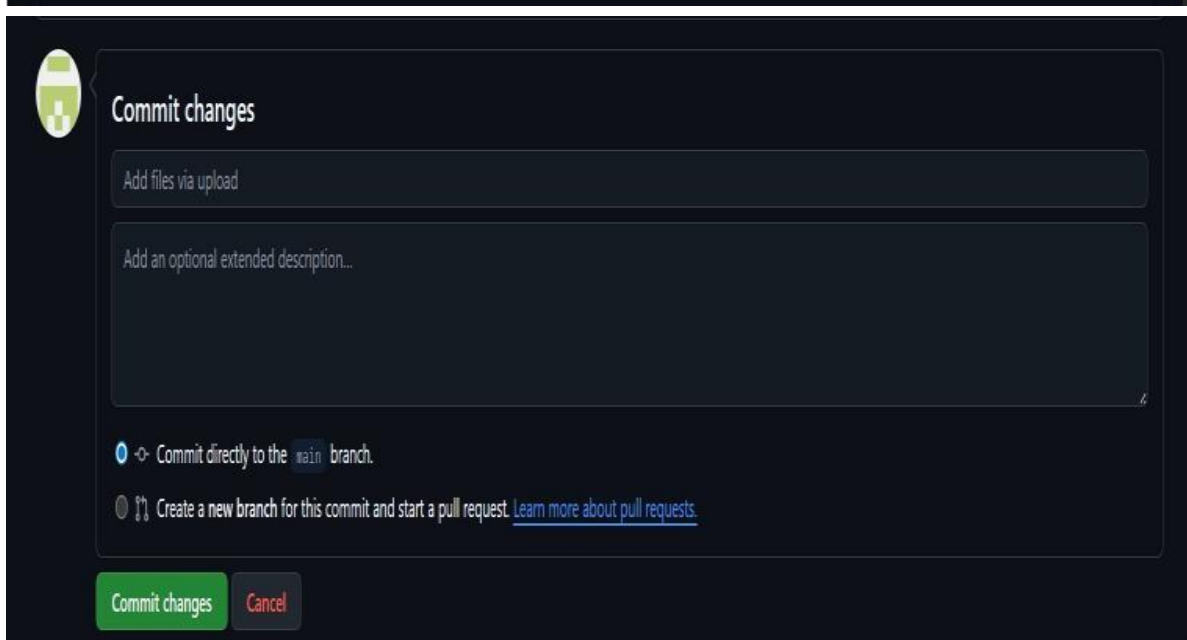
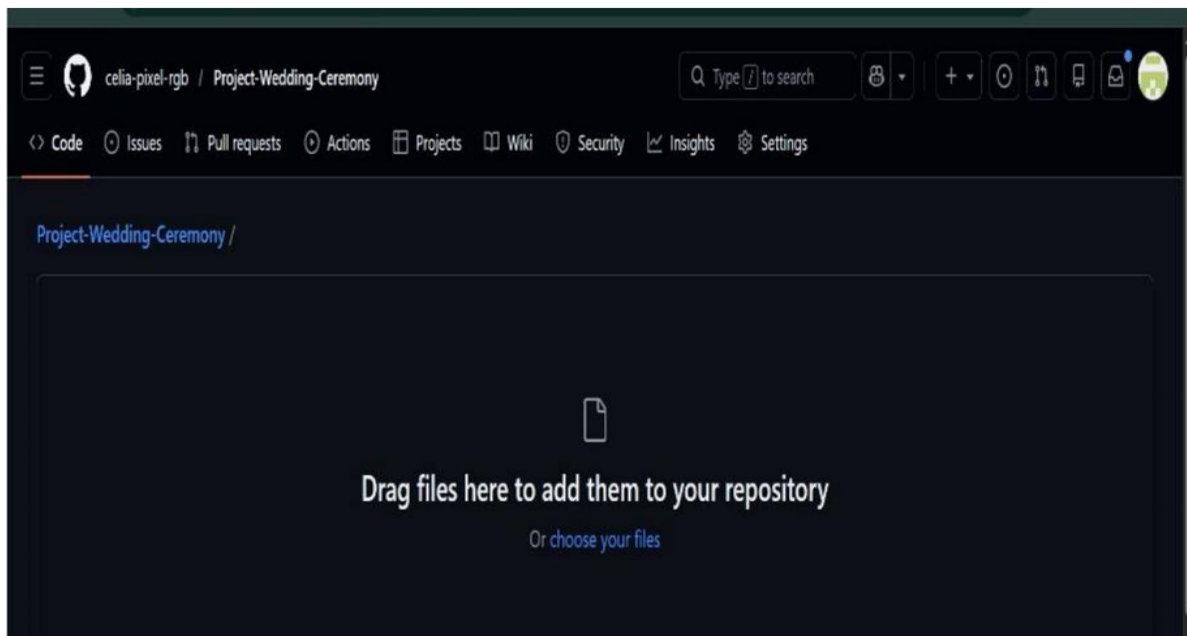
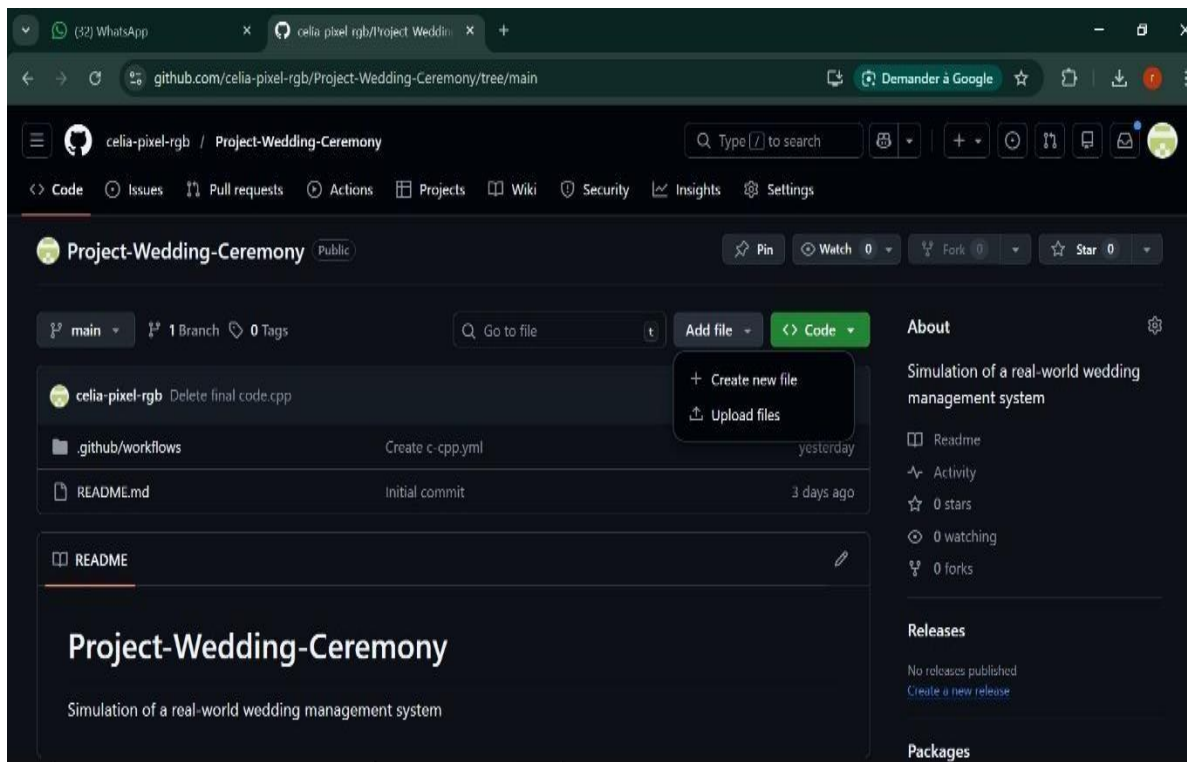
3.3. Cloning the Repository and first Push

After accepting their invitation, each member cloned the repository into their local machine. The group leader then performed the first push of the program.

```
Microsoft Windows [version 10.0.17763.1577]
(c) 2018 Microsoft Corporation. Tous droits réservés.

C:\Users\user>git clone https://github.com/celia-pixel-rgb/Project-Wedding-Ceremony.git
Cloning into 'Project-Wedding-Ceremony'...
remote: Enumerating objects: 4, done.
remote: Counting objects: 100% (4/4), done.
remote: Compressing objects: 100% (4/4), done.
remote: Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (4/4), done.

C:\Users\user>
```

4. Work Organisation and task distribution

The group consisted of 5 members. The work organisation depended on assigning responsibilities with respect to skills and the project phases. This approach permitted us to ensure that each section of this report would benefit from dedicated expertise.

Group Member	Main Role	Assigned Tasks
1. Matchim Dzuyim Verenne Celia	Implementation and testing	Common tasks: Project Overview, Development Tools Used, System Architecture, Phase 1 Summary. Individual tasks: GTK, CSV, Test of Person Management, Test of Category Management.
2. Mekolle Tuatsop Dalhia Aure	Documentation and planning	Common tasks: Project Overview, Development Tools Used, System Architecture, Phase 1 Summary. Individual tasks: Report 1, Gantt Project, Report 2, Implementation of Category Management.
3. Talla Tchossi Daniella Elsa	Development (core modules)	Common tasks: Project Overview, Development Tools Used, System Architecture, Phase 1 Summary. Individual tasks: GTK, CSV, Implementation of Person Management.
4. Messi Zang Paul Jacques	System design (UML diagrams)	Common tasks: Project Overview, Development Tools Used, System Architecture, Phase 1 Summary. Individual tasks: Use Case Diagram, Class Diagram.
5. Djomo Yepdi Evans	System modeling and deployment	Common tasks: Project Overview, Development Tools Used, System Architecture, Phase 1 Summary. Individual tasks: Sequence Diagram, Deployment Diagram.

Table 3: Work organisation and task distribution

5. System Architecture

The application follows a **simplified modular architecture inspired by the MVC design pattern**. The architecture consists of three main components:



Model

The model contains the core data structures and logic of the program, including:

- Person management
- Category management
- Priority management
- Gift management



Controller

The controller handles the flow of the application by managing the interactions between the user interface and the program modules.



View

The view represents the interface presented to the user. In this project, the view is implemented through a **console-based menu interface** that allows the user to navigate through the different system functionalities.

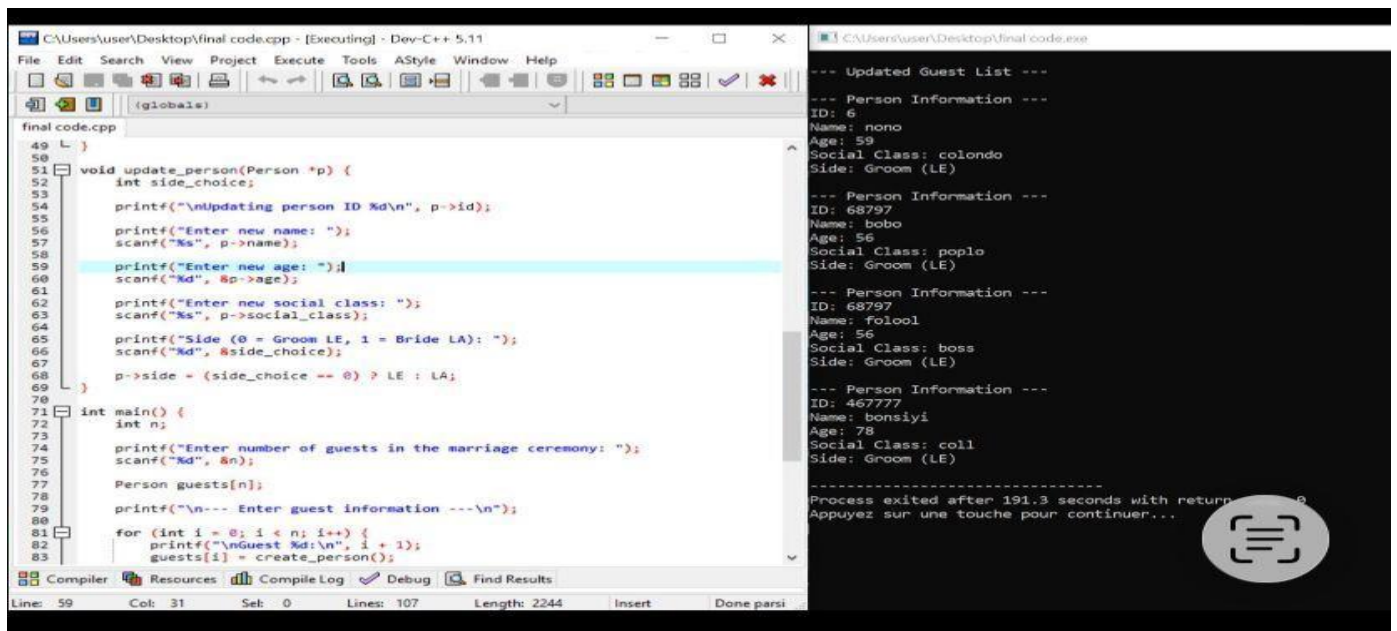
This architecture ensures clear separation between data, logic, and user interaction.

6. Functional Modules of the Application

As first stage, we did what we consider as our Module 1: Personal Management. This module focuses on:

6.1. Create a new guest

Here, we are creating a program that can create persons (guests). Based on the wanted total number of guests entered by the user, we defined a structure in which we register guests base on their name, age, Id, social status and their side (either the bride or the



```
49 }
50
51 void update_person(Person *p) {
52     int side_choice;
53
54     printf("\nUpdating person ID %d\n", p->id);
55     printf("Enter new name: ");
56     scanf("%s", p->name);
57
58     printf("Enter new age: ");
59     scanf("%d", &p->age);
60
61     printf("Enter new social class: ");
62     scanf("%s", p->social_class);
63
64     printf("Side (0 = Groom LE, 1 = Bride LA): ");
65     scanf("%d", &side_choice);
66
67     p->side = (side_choice == 0) ? LE : LA;
68 }
69
70 int main() {
71     int n;
72
73     printf("Enter number of guests in the marriage ceremony: ");
74     scanf("%d", &n);
75
76     Person guests[n];
77
78     printf("\n--- Enter guest information ---\n");
79
80     for (int i = 0; i < n; i++) {
81         printf("\nGuest %d:\n", i + 1);
82         guests[i] = create_person();
83     }
84 }
```

```
--- Updated Guest List ---
--- Person Information ---
ID: 6
Name: nono
Age: 59
Social Class: colondo
Side: Groom (LE)
--- Person Information ---
ID: 68797
Name: bobo
Age: 56
Social Class: poplo
Side: Groom (LE)
--- Person Information ---
ID: 68797
Name: folool
Age: 56
Social Class: boss
Side: Groom (LE)
--- Person Information ---
ID: 467777
Name: bonsiyi
Age: 78
Social Class: coll
Side: Groom (LE)
-----
Process exited after 191.3 seconds with return code 0
Appuyez sur une touche pour continuer...
```

groom).

6.2. Display the guest

```
void display_person(Person p) {  
    printf("\n--- Person Information ---\n");  
    printf("ID: %d\n", p.id);  
    printf("Name: %s\n", p.name);  
    printf("Age: %d\n", p.age);  
    printf("Social Class: %s\n", p.social_class);  
  
    if (p.side == LE)  
        printf("Side: Groom (LE)\n");  
    else  
        printf("Side: Bride (LA)\n");  
}
```

7. Main Difficulties Encountered

During the development of the project, several challenges were encountered:

- Understanding the use of **structures in C programming**
- Correctly implementing **alternatives structures, loops and memory management**
- Designing linked list structures for dynamic data
- Managing user input validation
- Debugging errors during program execution

These challenges required careful testing and collaboration among team members.

8. Solutions Implemented

To address the difficulties encountered, the following solutions were implemented:

- Reviewing C programming documentation and tutorials
- Testing each function independently before integration
- Using debugging techniques to identify program errors
- Collaborating through GitHub to track and fix problems

These solutions allowed the team to successfully implement the first module of the project.

9. Phase 1 Summary

The first phase of the project was successfully completed.

The team was able to:

- Understand the project requirements
- Set up the development environment
- Create the project structure
- Implement the **Person Management module**
- Test the system functionality

The project is now ready to proceed to the next phase, which will involve implementing additional modules that we will present later on.

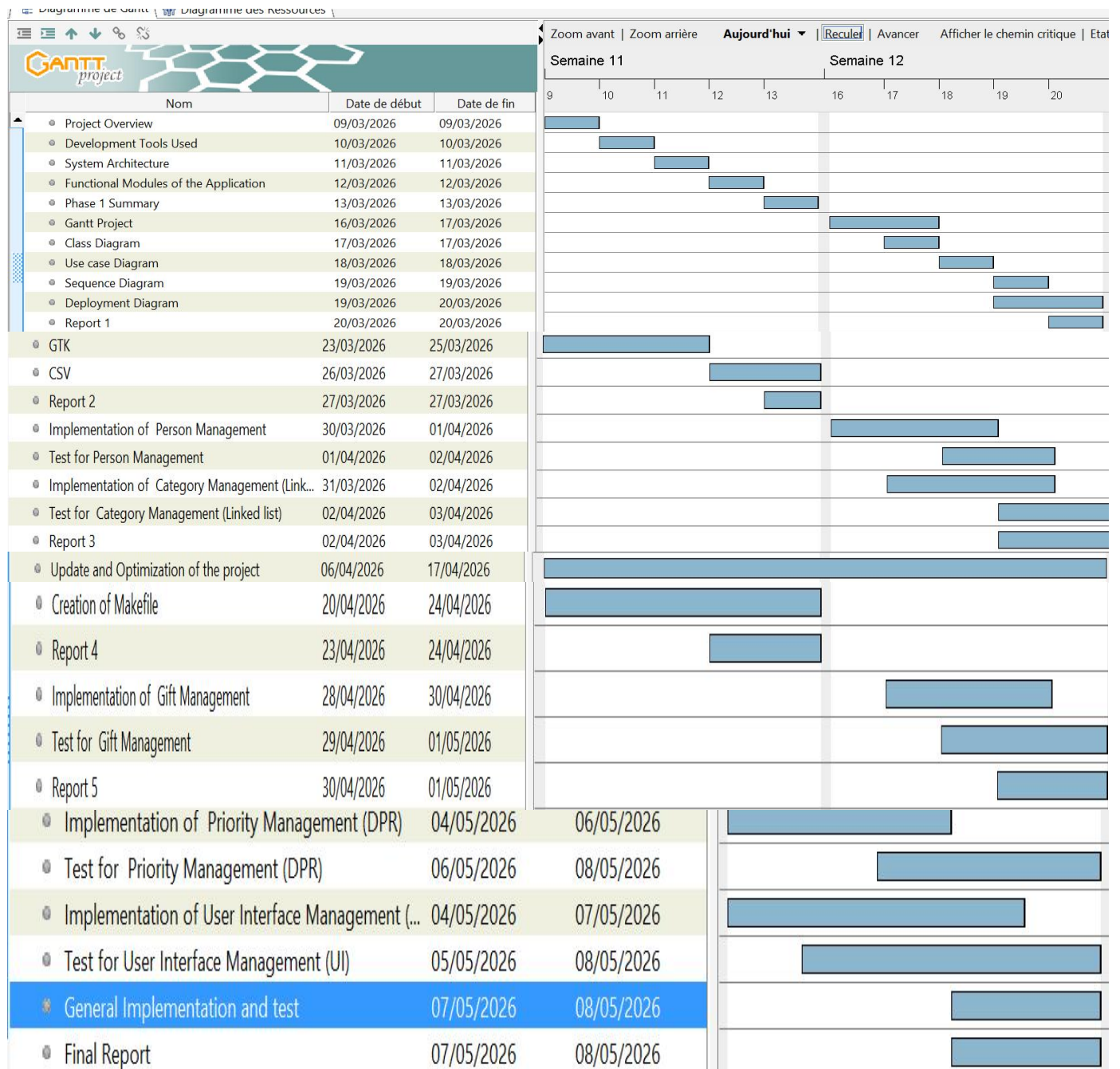
10. References

The following references were used during the development of this project:

- **Brian W. Kernighan & Dennis M. Ritchie** – *The C Programming Language*
- **Stephen Prata** – *C Programming: A Modern Approach*
- **ISO C Programming Documentation**
- **GitHub Official Documentation** – <https://docs.github.com>
- Course materials provided by the lecturer

11. Gantt Project

This project outlines the planning, design, and implementation of a software system using structured project management techniques. The Gantt chart was used to organize tasks, timelines, and dependencies efficiently.



The project was successfully structured using a Gantt chart, ensuring proper time management and task coordination. All phases were completed systematically, resulting in a functional and well-tested application.

12. SYSTEM DESIGN

System design is a critical phase in software development that defines the structure, components, and interactions of a system. It provides a clear blueprint to guide the development process and ensure that all functional and technical requirements are met.

In this section, we will present and explain the key design diagrams used in the project, including the class diagram, sequence diagram, deployment diagram, and use case diagram, to illustrate how the system operates and how its components interact.

12.1 CLASS DIAGRAM

The WIGMS UML Class Diagram models wedding organization entities using StarUML. A central Wedding class connects nine supporting classes — each with detailed attributes and defined relationships via UML associations and multiplicity notation.

- A class diagram provides a static view of the system, showing classes, attributes, operations, and relationships.
- In WIGMS, it acts as a blueprint for implementation.
- There are four classes; person, parking, gift and category

a. Class Diagram

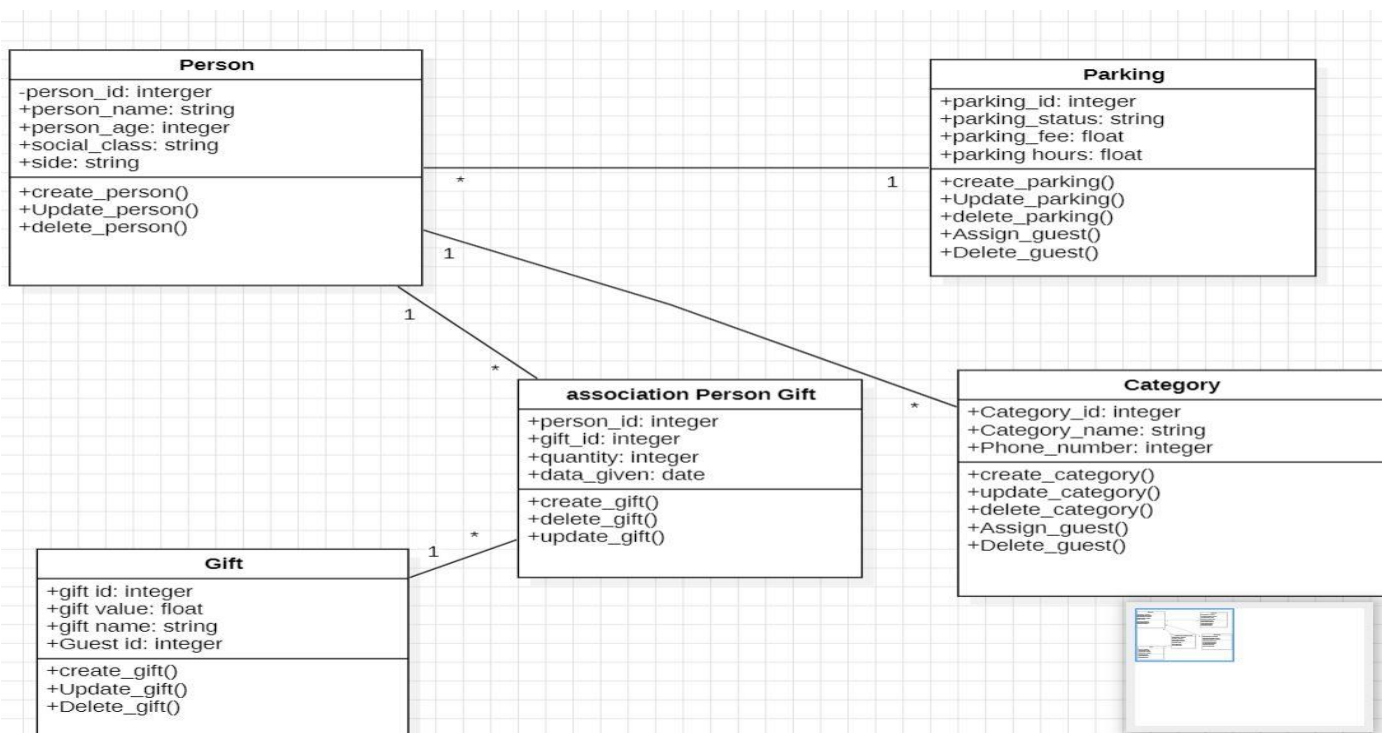


Figure 1: Class Diagram — Wedding Management System (StarUML)

Class	Attributes (private -)	Operations (public +)	Relationship
Person	id, name, age, social_class, side	create, update, delete	* → 1 Parking* → 1 Category* ↔ * Gift
Parking	id, status, fee	create, update	1 → * Person
Gift	id, value, name, guest_id	create, update, delete	* → 1 Category* ↔ * Person
Category	id, name, phone_number	create, update, delete	1 → * Person 1 → * Gift

Table 4: System classes, attributes, and relationships

12.2 USE CASE DIAGRAM

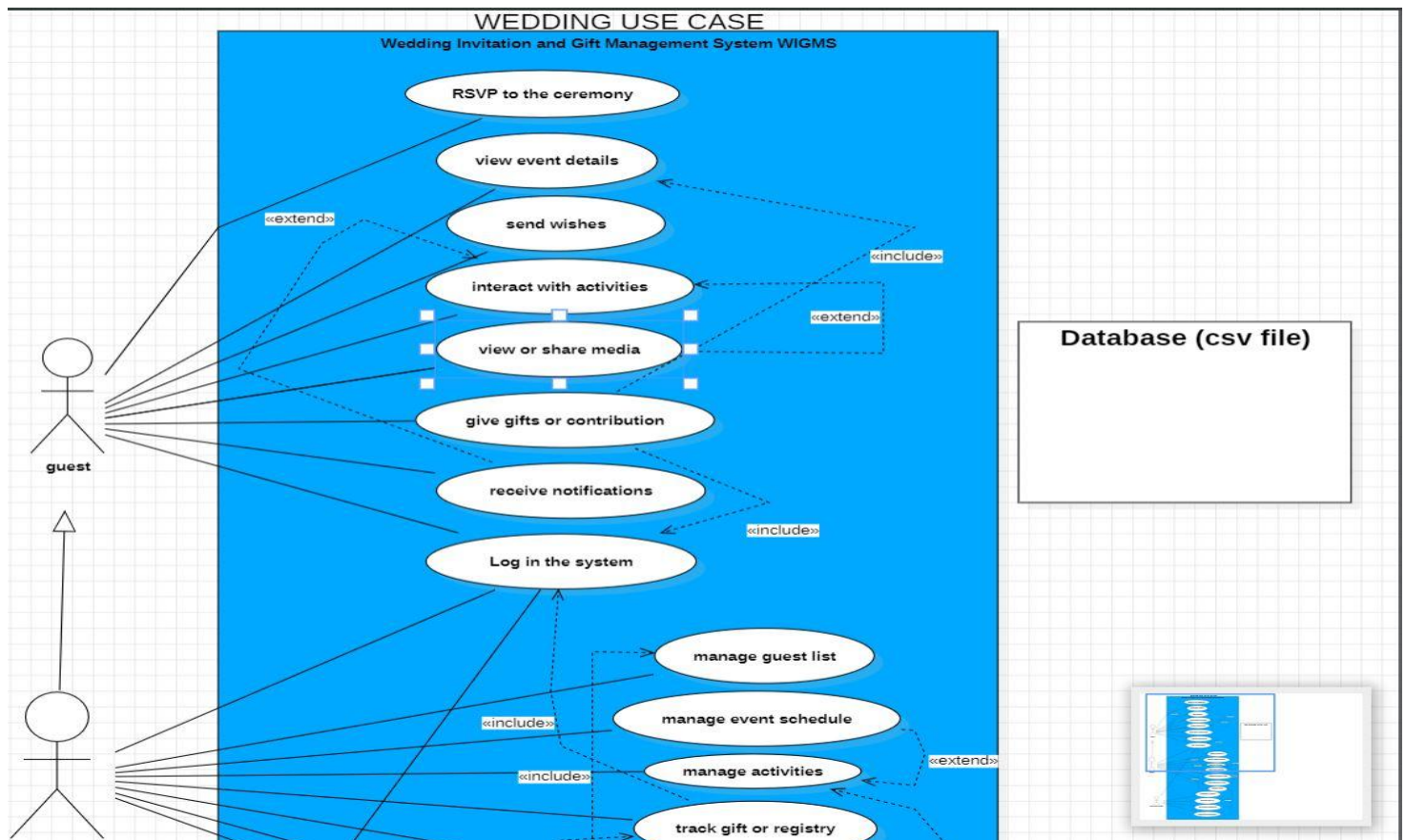


Figure 2: Usecase Diagram — Wedding Management System (StarUML)

12.3 SEQUENCE DIAGRAM

Overview

A sequence diagram illustrates how different components of the wedding-planning system interact over time through ordered message exchanges.

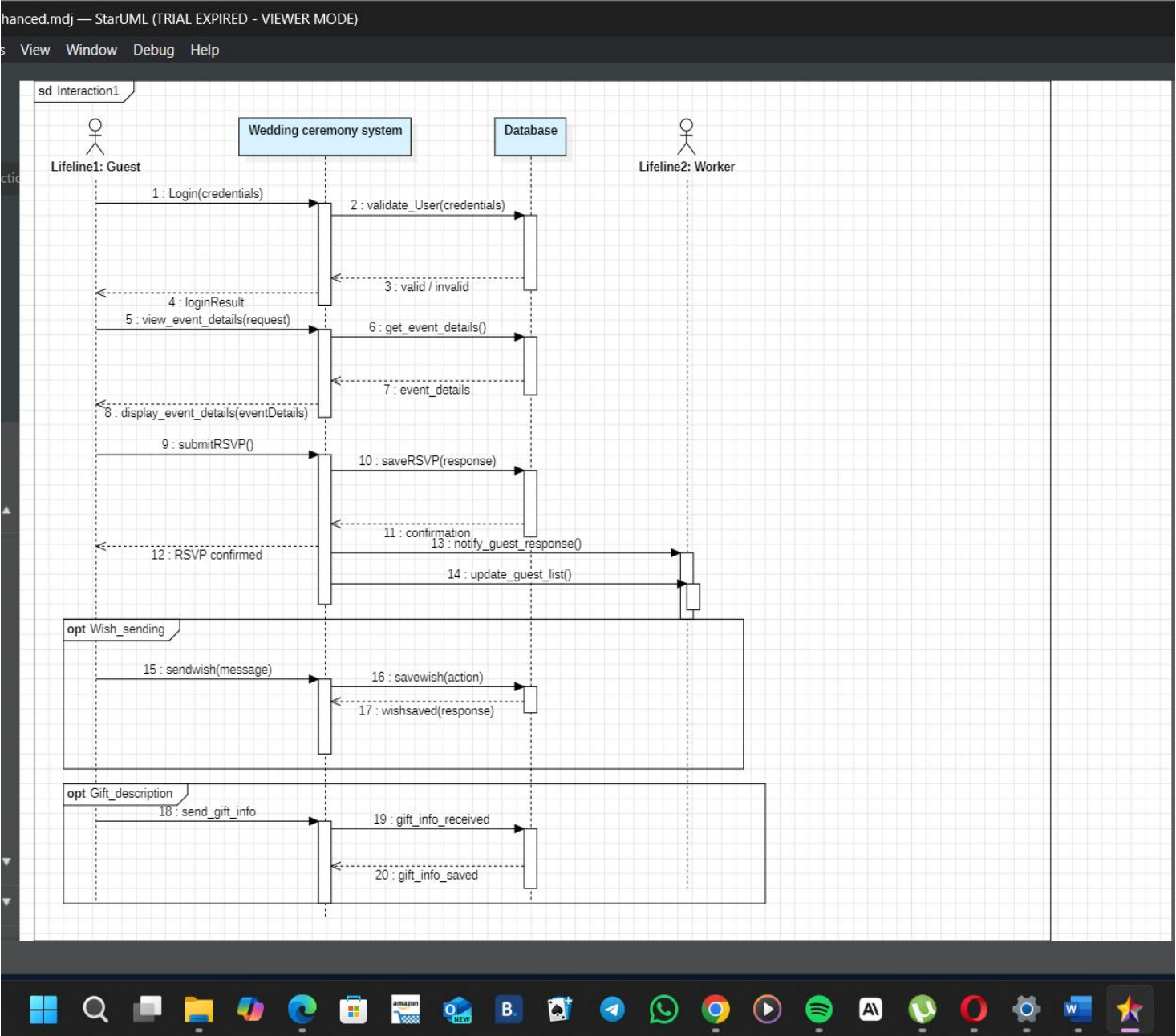


Figure 3: Sequence Diagram — Wedding Management System (StarUM

It helps visualize the workflow of actions such as event creation, guest invitations, and RSVP tracking, ensuring processes occur in the correct sequence.

12.4 MNEMONIC TABLE

This table will enable any person familiar with c-programming projects who may or may have not participated in the project to better understand the actions going on represented on both the use case and sequence diagrams already illustrated above.

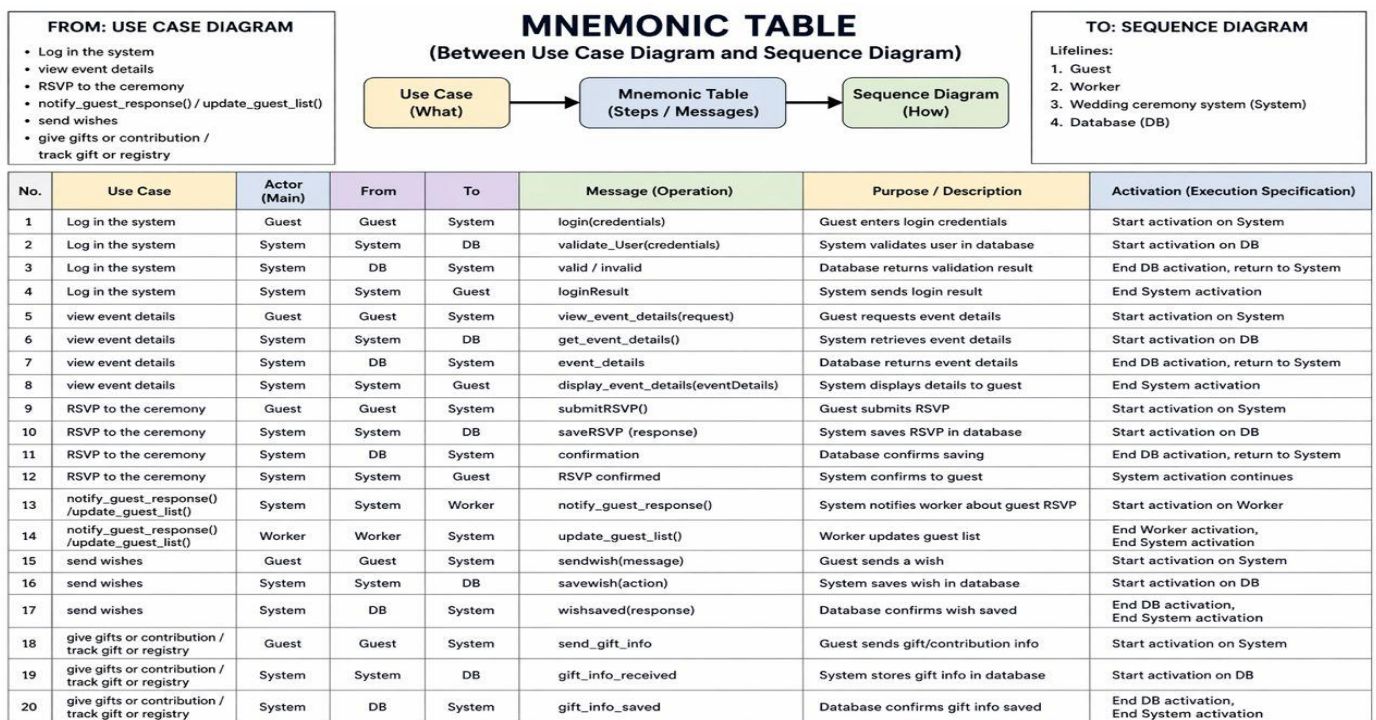


Figure 5: Mnemonic table between the usecase diagram and sequence diagram — Wedding Management System (StarUML)

12.5 DEPLOYMENT DIAGRAM

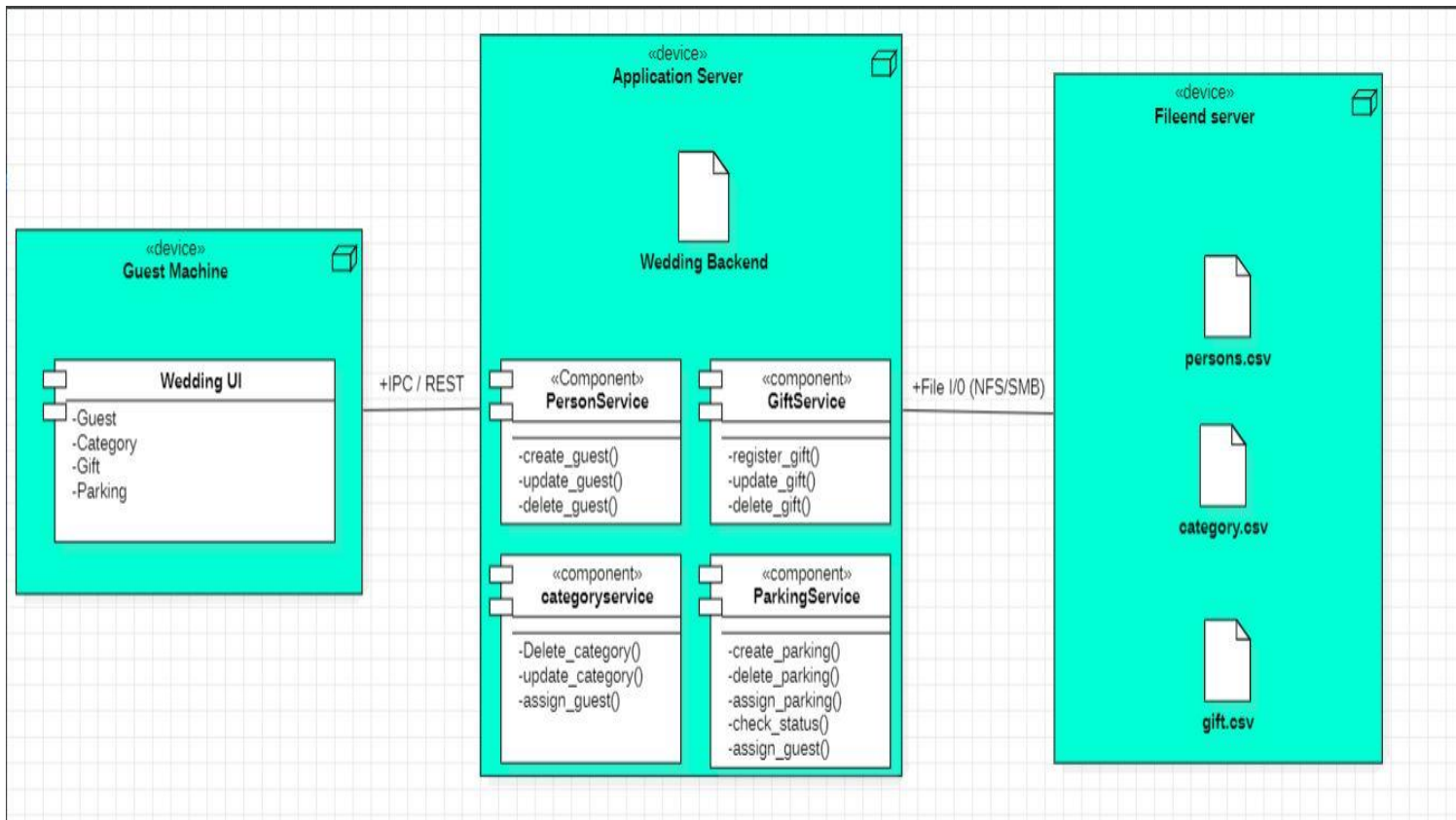


Figure 4: Deployment Diagram — Wedding Management System (StarUML)

13. GNU Image Manipulation Program Toolkit (GTK) and Comma-Selected Value (CSV

13.1 GTK

Development of a GTK-based form application written in C for managing wedding guest data. The application collects user input through a graphical interface and stores it in a CSV file.

Program Overview

- **Download and Installation of GTK 4 for windows systems**

The program uses GTK4 to create a graphical interface consisting of input fields for Name, Age, Status,

First, you have to download the installer at msys2.org, selecting the first version for Windows.

1. Download the installer:

 [msys2-x86_64-20260322.exe](#)

or

 [msys2-arm64-20260322.exe](#)

(ARM64)

(Optional) For more information on the installer, like command line options, or how to verify the checksum and signature of the installer, see the [installer guide](#).

- Following this, at the end of the download process, ensure that you store the installer in a folder you will easily remember e.g wedding management ceremony.
- Again, open the “MSYS2 MSYS” terminal and run the command “pacman -Syu”, and press “Y” until you finalize the installations.

```
user@DESKTOP-S24HAFG MSYS ~
$ pacman -Syu
:: Synchronizing package databases...
clangarm64                    521.3 KiB  12.4 KiB/s  00:42 [#####] 100%
mingw32                      110.7 KiB   7.34 KiB/s  00:15 [#####] 100%
mingw64                      459.4 KiB  21.8 KiB/s  00:21 [#####] 100%
ucrt64                       538.8 KiB  13.1 KiB/s  00:41 [#####] 100%
clang64                      529.2 KiB   9.69 KiB/s  00:55 [#####] 100%
msys                         456.2 KiB  52.0 KiB/s  00:09 [#####] 100%
```

- Still, run the command “\$ pacman -S mingw-w64-ucrt-x86_64-gcc” into the “MSYS2 UCR64” terminal, and validate all the options till the end of the various installations.

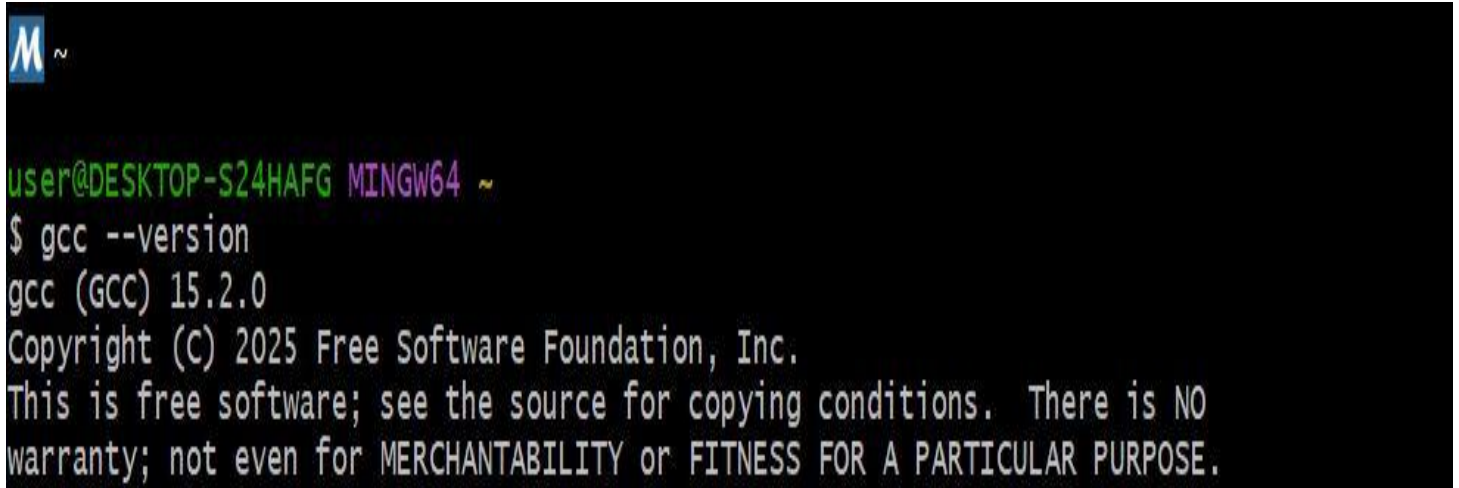
```
user@DESKTOP-S24HAFG UCR64 ~
$ pacman -S mingw-w64-ucrt-x86_64-gcc
resolving dependencies...
looking for conflicting packages...

Packages (2) mingw-w64-ucrt-x86_64-gcc-libs-15.2.0-14  mingw-w64-ucrt-x86_64-gcc-15.2.0-14

Total Download Size:    47.67 MiB
Total Installed Size:  248.34 MiB
Net Upgrade Size:        0.07 MiB

:: Proceed with installation? [Y/n]
:: Retrieving packages...
mingw-w64-ucrt-x86_64-gcc-...  546.8 KiB  23.4 KiB/s  33:34 [-----] 1%
mingw-w64-ucrt-x86_64-gcc-...  804.7 KiB  40.2 KiB/s  00:06 [#####] 76%
Total (0/2)                  1351.5 KiB  73.2 KiB/s  10:48 [-----] 2%
```

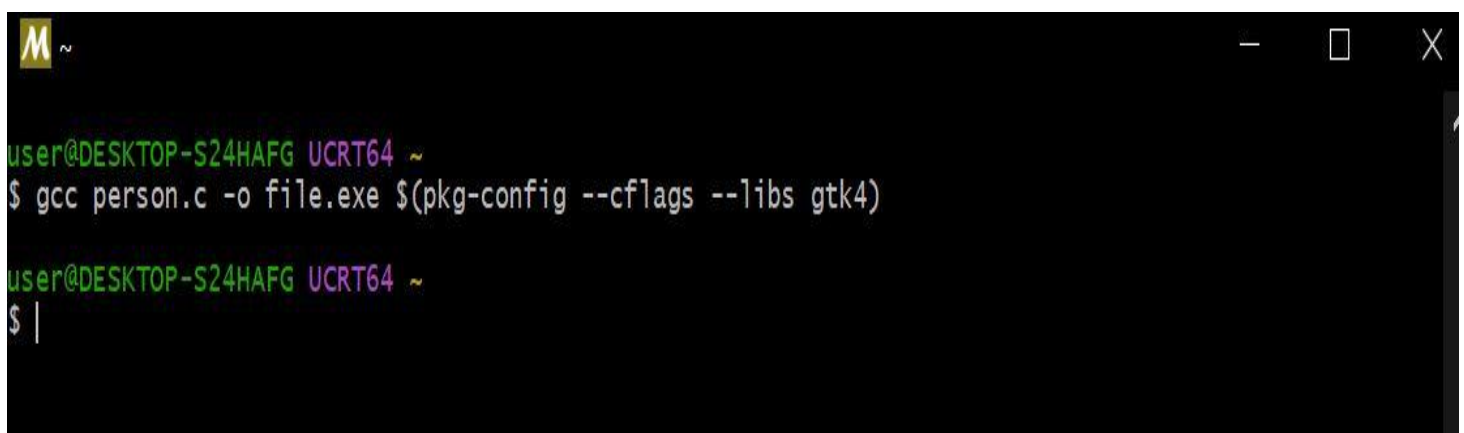
- To continue, install GTK still in the “MSYS2 MINGW64” terminal by typing the command “`pacman -S mingw-w64-x86_64-gtk4`”, and then install build tools by typing-in the command “`pacman -S mingw-w64-x86_64-pkgconfig mingw-w64-x86_64-make`” in the same terminal.
- Next, in order to ensure that the gcc compiler was well installed, run the command “`gcc --version`” into the “MSYS2 MINGW64” terminal. You should see the version afterwards.

A terminal window with a black background and a blue 'M' icon in the top left corner. The prompt is 'user@DESKTOP-S24HAFG MINGW64 ~'. The command '\$ gcc --version' has been entered, and the output is displayed in green text: 'gcc (GCC) 15.2.0', 'Copyright (C) 2025 Free Software Foundation, Inc.', and 'This is free software; see the source for copying conditions. There is NO warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.'

- You have now installed your GTK library, you should proceed further with the execution of codes.

NB

- ✓ Before running a code, you must make sure that it is stored in the same folder as the one containing your GTK library files.
- ✓ If the code ran successfully, a “.exe” file appears just below your code in the folder.
 - In order to run the code, you must open the “MSYS UCR64”

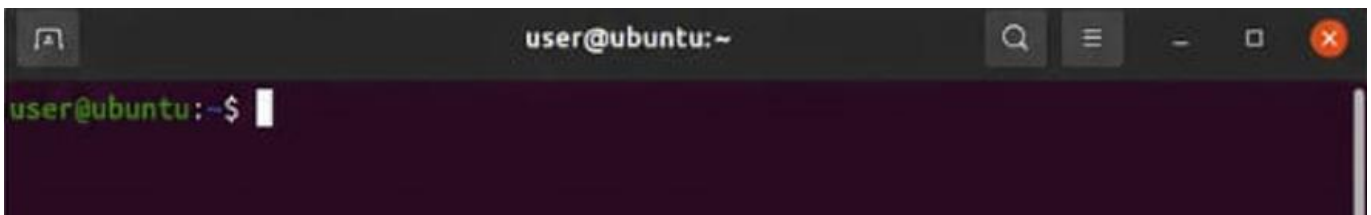
A terminal window with a black background and a yellow 'M' icon in the top left corner. The prompt is 'user@DESKTOP-S24HAFG UCRT64 ~'. The command '\$ gcc person.c -o file.exe \$(pkg-config --cflags --libs gtk4)' has been entered. The prompt is now '\$ |'.

terminal and type the command “gcc file.c -o file.exe \$(pkg-config --cflags --libs gtk4)”

- Download and installation of GTK for Linux

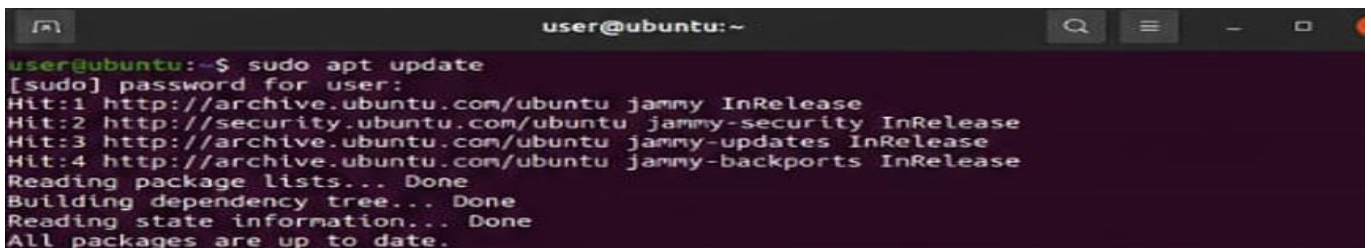
To install GTK for some particular systems as “Linux”, we follow different steps:

➤ Step1: Open the terminal



➤ Step2: Update your system

Here, you run the following command “*sudo apt update*” to update the package list.



➤ Step3: install GTK

You run the following command “*sudo apt install libgtk-4-dev*” in order to install the GTK4 development packages



- After entering the command, you need to confirm the installation by “typing Y and press Enter”. Therefore, the packages will be downloaded and installed.

```
Get:1 http://archive.ubuntu.com/ubuntu jammy/main amd64 libgtk-4-1 amd64 4.10.2-1ubuntu1 [2,598 kB]
Get:2 http://archive.ubuntu.com/ubuntu jammy/main amd64 gir1.2-gtk-4.0 amd64 4.10.2-1ubuntu1 [107 kB]
Get:3 http://archive.ubuntu.com/ubuntu jammy/main amd64 libgtk-4-dev amd64 4.10.2-1ubuntu1 [4,110 kB]
Fetched 6,815 kB in 2s (3,329 kB/s)
Selecting previously unselected package libgtk-4-1:amd64.
(Reading database ... 197334 files and directories currently installed.)
Preparing to unpack .../libgtk-4-1_4.10.2-1ubuntu1_amd64.deb ...
Unpacking libgtk-4-1:amd64 (4.10.2-1ubuntu1) ...
Selecting previously unselected package gir1.2-gtk-4.0:amd64.
Preparing to unpack .../gir1.2-gtk-4.0_4.10.2-1ubuntu1_amd64.deb ...
Unpacking gir1.2-gtk-4.0:amd64 (4.10.2-1ubuntu1) ...
Selecting previously unselected package libgtk-4-dev:amd64.
Preparing to unpack .../libgtk-4-dev_4.10.2-1ubuntu1_amd64.deb ...
Unpacking libgtk-4-dev:amd64 (4.10.2-1ubuntu1) ...
Setting up libgtk-4-1:amd64 (4.10.2-1ubuntu1) ...
Setting up gir1.2-gtk-4.0:amd64 (4.10.2-1ubuntu1) ...
Setting up libgtk-4-dev:amd64 (4.10.2-1ubuntu1) ...
Processing triggers for libc-bin (2.35-0ubuntu3.1) ...
user@ubuntu:~$
```

➤ *Step4: Verifying the installation*

After installing, you need to check the installed GTK4 version. To do it, you enter the command “*pkg-config --modversion gtk4*”

```
user@ubuntu:~$ pkg-config --modversion gtk4
4.10.2
user@ubuntu:~$
```

NB: In order to verify the installed GTK4 and its functionality, you can decide to create and test a program. To compile the program, you need to enter the following command

“*gcc name_of_the_program.c `pkg-config --cflags --libs gtk4` -o name_of_the_program*”

Then to test, we run: “*./name_of_the_program*”

EX:

```
user@ubuntu:~$ gcc test.c `pkg-config --cflags --libs gtk4` -o test
user@ubuntu:~$ ./test
user@ubuntu:~$
```

14. Key Implementation Details

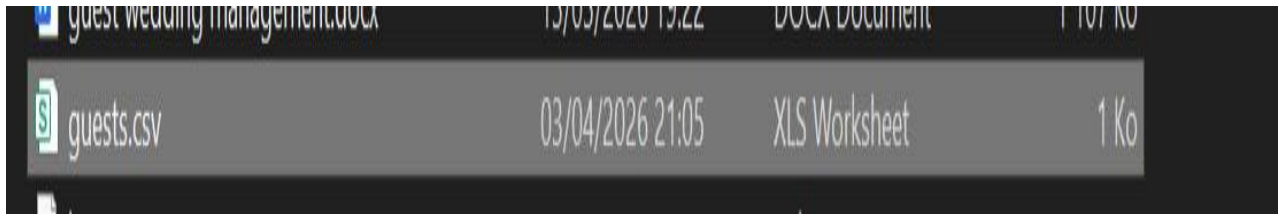
```
marriage.c  x
1  #include <gtk/gtk.h>
2  #include <stdio.h>
3
4  // Input fields (global so we can access them easily)
5  GtkWidget *name_entry;
6  GtkWidget *age_entry;
7  GtkWidget *status_entry;
8  GtkWidget *phone_entry;
9  GtkWidget *side_entry;
10
11 // Function called when button is clicked
12 void save_data(GtkButton *button, gpointer user_data) {
13
14     // Get text from input fields
15     const char *name = gtk_editable_get_text(GTK_EDITABLE(name_entry));
16     const char *age = gtk_editable_get_text(GTK_EDITABLE(age_entry));
17     const char *status = gtk_editable_get_text(GTK_EDITABLE(status_entry));
18     const char *phone = gtk_editable_get_text(GTK_EDITABLE(phone_entry));
19     const char *side = gtk_editable_get_text(GTK_EDITABLE(side_entry));
20
21     // Open CSV file
22     FILE *file = fopen("guests.csv", "a");
23
24     if (file == NULL) {
25         g_print("Error opening file\n");
26         return;
27     }
28
29     // Write data into file
30     fprintf(file, "%s,%s,%s,%s,%s\n", name, age, status, phone, side);
31
32     fclose(file);
33
34     g_print("Data saved!\n");
35 }
36
37 // Function that creates the window
38 void activate(GtkApplication *app, gpointer user_data) {
39     GtkWidget *window;
40
41     // Function that creates the window
42     void activate(GtkApplication *app, gpointer user_data) {
43
44         GtkWidget *window;
45         GtkWidget *grid;
46         GtkWidget *button;
47
48         // Create window
49         window = gtk_application_window_new(app);
50         gtk_window_set_title(GTK_WINDOW(window), "Wedding Form");
51         gtk_window_set_default_size(GTK_WINDOW(window), 400, 300);
52
53         // Create Layout
54         grid = gtk_grid_new();
55         gtk_window_set_child(GTK_WINDOW(window), grid);
56
57         // Create input fields
58         name_entry = gtk_entry_new();
59         age_entry = gtk_entry_new();
60         status_entry = gtk_entry_new();
61         phone_entry = gtk_entry_new();
62         side_entry = gtk_entry_new();
63
64         // Add Labels and fields
65         gtk_grid_attach(GTK_GRID(grid), gtk_label_new("Name"), 0, 0, 1, 1);
66         gtk_grid_attach(GTK_GRID(grid), name_entry, 1, 0, 1, 1);
67
68         gtk_grid_attach(GTK_GRID(grid), gtk_label_new("Age"), 0, 1, 1, 1);
69         gtk_grid_attach(GTK_GRID(grid), age_entry, 1, 1, 1, 1);
70
71         gtk_grid_attach(GTK_GRID(grid), gtk_label_new("Status"), 0, 2, 1, 1);
72         gtk_grid_attach(GTK_GRID(grid), status_entry, 1, 2, 1, 1);
73
74         gtk_grid_attach(GTK_GRID(grid), gtk_label_new("Phone"), 0, 3, 1, 1);
75         gtk_grid_attach(GTK_GRID(grid), phone_entry, 1, 3, 1, 1);
76
77         gtk_grid_attach(GTK_GRID(grid), gtk_label_new("Side"), 0, 4, 1, 1);
78         gtk_grid_attach(GTK_GRID(grid), side_entry, 1, 4, 1, 1);
79
80         // Create button
81         phone_entry = gtk_entry_new();
82         side_entry = gtk_entry_new();
83
84         // Add Labels and fields
85         gtk_grid_attach(GTK_GRID(grid), gtk_label_new("Name"), 0, 0, 1, 1);
86         gtk_grid_attach(GTK_GRID(grid), name_entry, 1, 0, 1, 1);
87
88         gtk_grid_attach(GTK_GRID(grid), gtk_label_new("Age"), 0, 1, 1, 1);
89         gtk_grid_attach(GTK_GRID(grid), age_entry, 1, 1, 1, 1);
90
91         gtk_grid_attach(GTK_GRID(grid), gtk_label_new("Status"), 0, 2, 1, 1);
92         gtk_grid_attach(GTK_GRID(grid), status_entry, 1, 2, 1, 1);
93
94         gtk_grid_attach(GTK_GRID(grid), gtk_label_new("Phone"), 0, 3, 1, 1);
95         gtk_grid_attach(GTK_GRID(grid), phone_entry, 1, 3, 1, 1);
96
97         gtk_grid_attach(GTK_GRID(grid), gtk_label_new("Side"), 0, 4, 1, 1);
98         gtk_grid_attach(GTK_GRID(grid), side_entry, 1, 4, 1, 1);
99
100        // Create button
101        button = gtk_button_new_with_label("Save");
102        gtk_grid_attach(GTK_GRID(grid), button, 0, 5, 2, 1);
103
104        // Connect button click to function
105        g_signal_connect(button, "clicked", G_CALLBACK(save_data), NULL);
106
107        gtk_window_present(GTK_WINDOW(window));
108    }
109
110    // Main function
111    int main(int argc, char **argv) {
112        GtkApplication *app;
113
114        app = gtk_application_new("org.example.wedding", G_APPLICATION_DEFAULT_FLAGS);
115        g_signal_connect(app, "activate", G_CALLBACK(activate), NULL);
116
117        return g_application_run(G_APPLICATION(app), argc, argv);
118    }
119 }
```

- ✓ **Global widget pointers** were used to allow access to input fields across functions. This simplifies retrieving user input when the Save button is clicked.
- ✓ **The use of `gtk_editable_get_text()` allows** extraction of user input from each entry field. This is required because GTK widgets do not directly expose their content.
- ✓ **File handling** is implemented using `fopen` in append mode ("a").
This ensures that new data is added without overwriting existing entries.

14.2 Csv file creation

The csv file was generated immediately after the GTK form worked, and was tested by inputting arbitrary entries

Person csv file



A	B	C	D
0,Matchim,16,VIP,679611233,0,Yes			

14.3 Problem Encountered and Solution

General Problem

- Difficulty in configuring development tools such as GTK.
- Limited understanding of UML diagrams (sequence, deployment, use case, class).
- Challenges in using StarUML (navigation, relationships, layout).
- Errors in coding such as file handling, compilation, and signal connections.
- Complexity in organizing system interactions and component communication.

14.4 General Solution

- Proper installation and configuration of development tools.
- Studying UML concepts and applying correct diagram rules.
- Practicing and exploring StarUML features for better usage.
- Applying debugging techniques and correct coding practices.
- Breaking the project into smaller tasks using structured planning tools like Gantt charts.

14.5 Conclusion

The Wedding Ceremony Management System demonstrates how C programming can be combined with project planning tools, system design diagrams, and user interface technologies. The use of GTK and CSV enhances usability and data management, making the system efficient and practical for real-world applications.

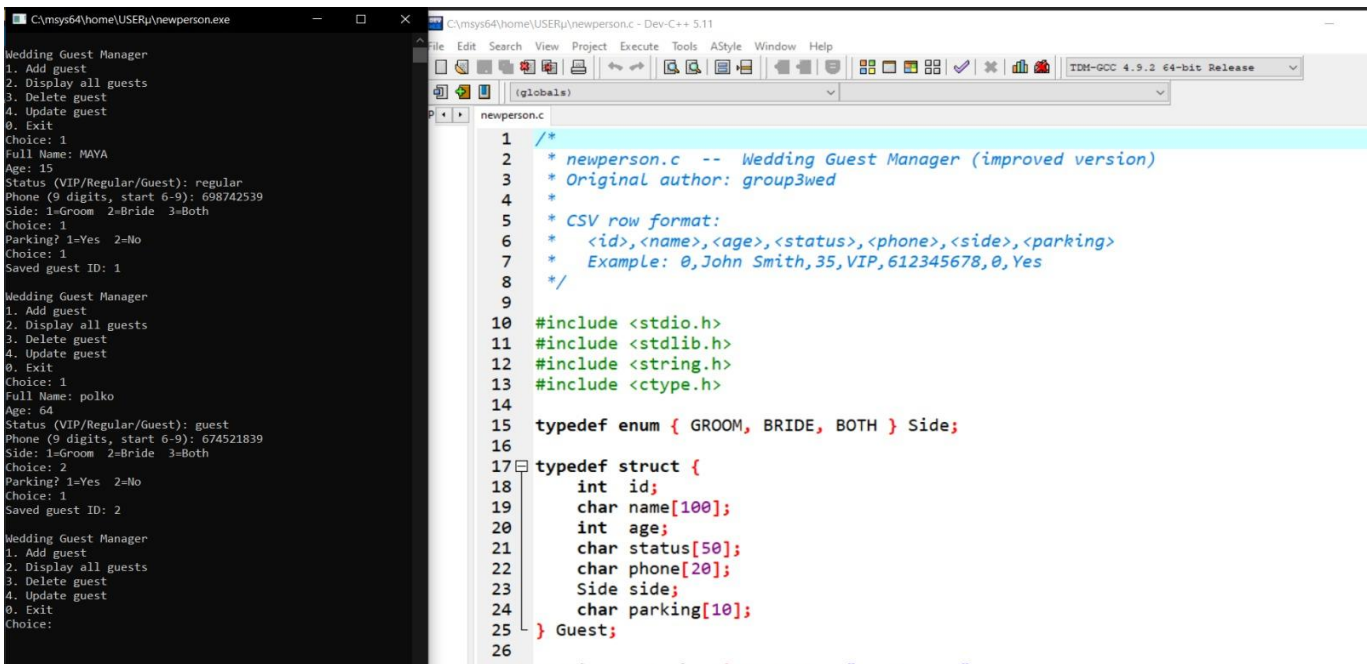
15. Person and category management: implementations & tests

15.1 Person Management

This module concerns the registration of the guests at a wedding ceremony, alongside all the operations carried out on them (Entry, Update, Delete).

- The previous model in the second report being lacking, the following is an updated version
- The c-program was written and thoroughly debugged using dev c++ software

15.1.1 Extract of Person Management C-code



```
Wedding Guest Manager
1. Add guest
2. Display all guests
3. Delete guest
4. Update guest
0. Exit
Choice: 1
Full Name: MAYA
Age: 15
Status (VIP/Regular/Guest): regular
Phone (9 digits, start 6-9): 698742539
Side: 1=Groom 2=Bride 3=Both
Choice: 1
Parking? 1=Yes 2=No
Choice: 1
Saved guest ID: 1

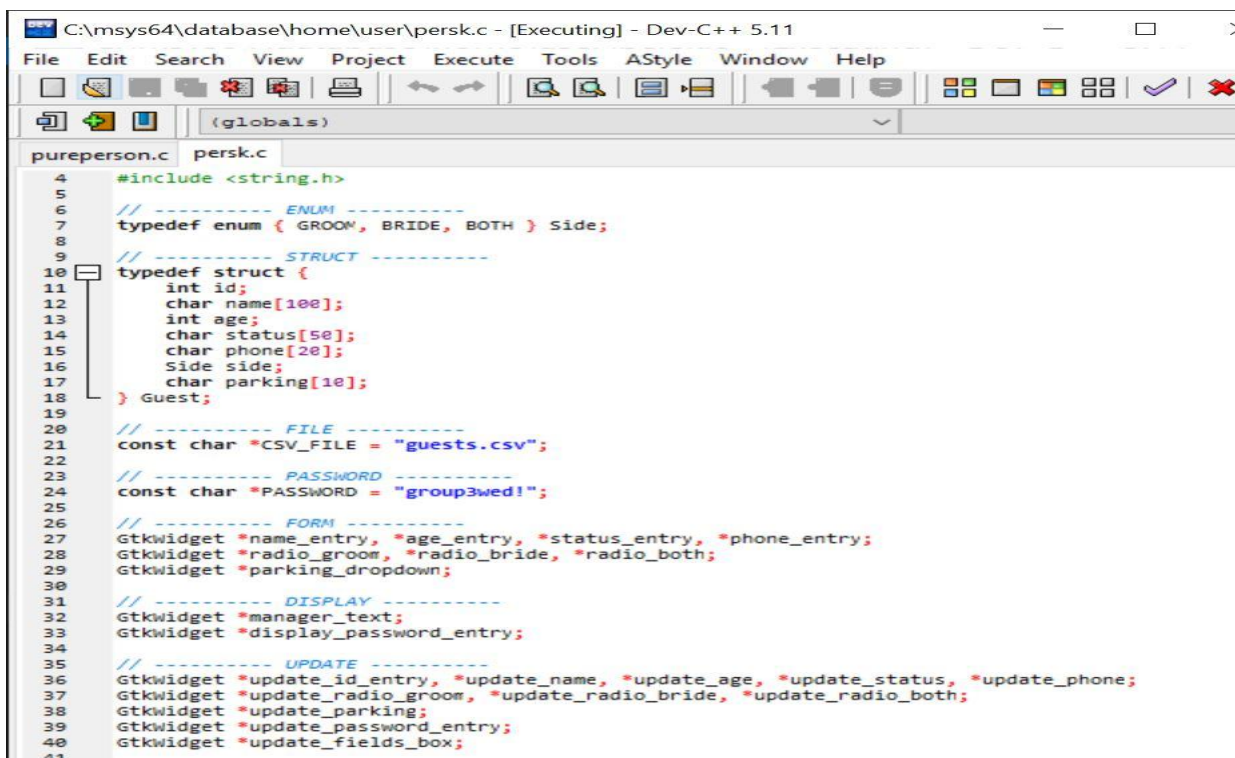
Wedding Guest Manager
1. Add guest
2. Display all guests
3. Delete guest
4. Update guest
0. Exit
Choice: 1
Full Name: polko
Age: 64
Status (VIP/Regular/Guest): guest
Phone (9 digits, start 6-9): 674521839
Side: 1=Groom 2=Bride 3=Both
Choice: 2
Parking? 1=Yes 2=No
Choice: 1
Saved guest ID: 2

Wedding Guest Manager
1. Add guest
2. Display all guests
3. Delete guest
4. Update guest
0. Exit
Choice:
```

```
1 /*
2  * newperson.c -- Wedding Guest Manager (improved version)
3  * Original author: group3wed
4  *
5  * CSV row format:
6  *   <id>,<name>,<age>,<status>,<phone>,<side>,<parking>
7  *   Example: 0,John Smith,35,VIP,612345678,0,Yes
8  */
9
10 #include <stdio.h>
11 #include <stdlib.h>
12 #include <string.h>
13 #include <ctype.h>
14
15 typedef enum { GROOM, BRIDE, BOTH } Side;
16
17 typedef struct {
18     int id;
19     char name[100];
20     int age;
21     char status[50];
22     char phone[20];
23     Side side;
24     char parking[10];
25 } Guest;
26
```

- Once written, the c-program was converted into a GTK form using GTK-specific libraries.

15.1.2 Extract of person management GTK c-code



```
4 #include <string.h>
5
6 // ----- ENUM -----
7 typedef enum { GROOM, BRIDE, BOTH } Side;
8
9 // ----- STRUCT -----
10 typedef struct {
11     int id;
12     char name[100];
13     int age;
14     char status[50];
15     char phone[20];
16     Side side;
17     char parking[10];
18 } Guest;
19
20 // ----- FILE -----
21 const char *CSV_FILE = "guests.csv";
22
23 // ----- PASSWORD -----
24 const char *PASSWORD = "group3wed!";
25
26 // ----- FORM -----
27 GtkWidget *name_entry, *age_entry, *status_entry, *phone_entry;
28 GtkWidget *radio_groom, *radio_bride, *radio_both;
29 GtkWidget *parking_dropdown;
30
31 // ----- DISPLAY -----
32 GtkWidget *manager_text;
33 GtkWidget *display_password_entry;
34
35 // ----- UPDATE -----
36 GtkWidget *update_id_entry, *update_name, *update_age, *update_status, *update_phone;
37 GtkWidget *update_radio_groom, *update_radio_bride, *update_radio_both;
38 GtkWidget *update_parking;
39 GtkWidget *update_password_entry;
40 GtkWidget *update_fields_box;
41
```

- The csv file was automatically generated below the gtk person management file immediately the gtk compiled successfully.

15.1.3 GTK ran successfully

```
user@DESKTOP-S24HAFG UCRT64 ~  
$ gcc updatedperson.c -o updatedperson.exe $(pkg-config --cflags --libs gtk4)  
  
user@DESKTOP-S24HAFG UCRT64 ~  
$ gcc persk.c -o persk.exe $(pkg-config --cflags --libs gtk4)  
  
user@DESKTOP-S24HAFG UCRT64 ~  
$ |
```

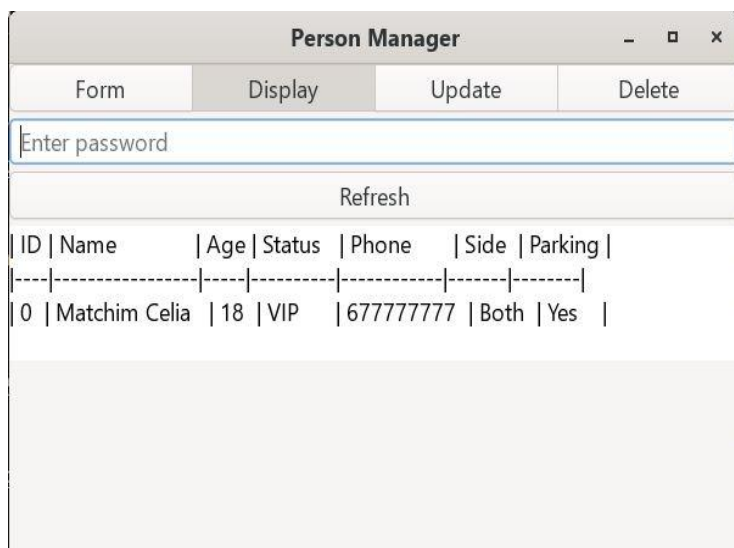
15.1.4 Executable File



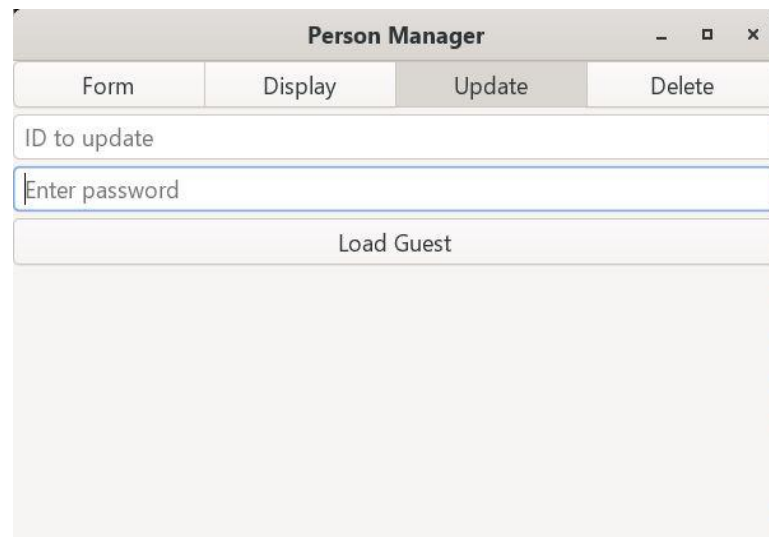
person.c	24/03/2026 20:15	C Source File	22 Ko
person.exe	01/04/2026 21:23	Application	148 Ko

- Tests were carried out to verify that the Person management form performs its job correctly

(a) Person registration



ID	Name	Age	Status	Phone	Side	Parking
0	Matchim Celia	18	VIP	677777777	Both	Yes



(b) Person update

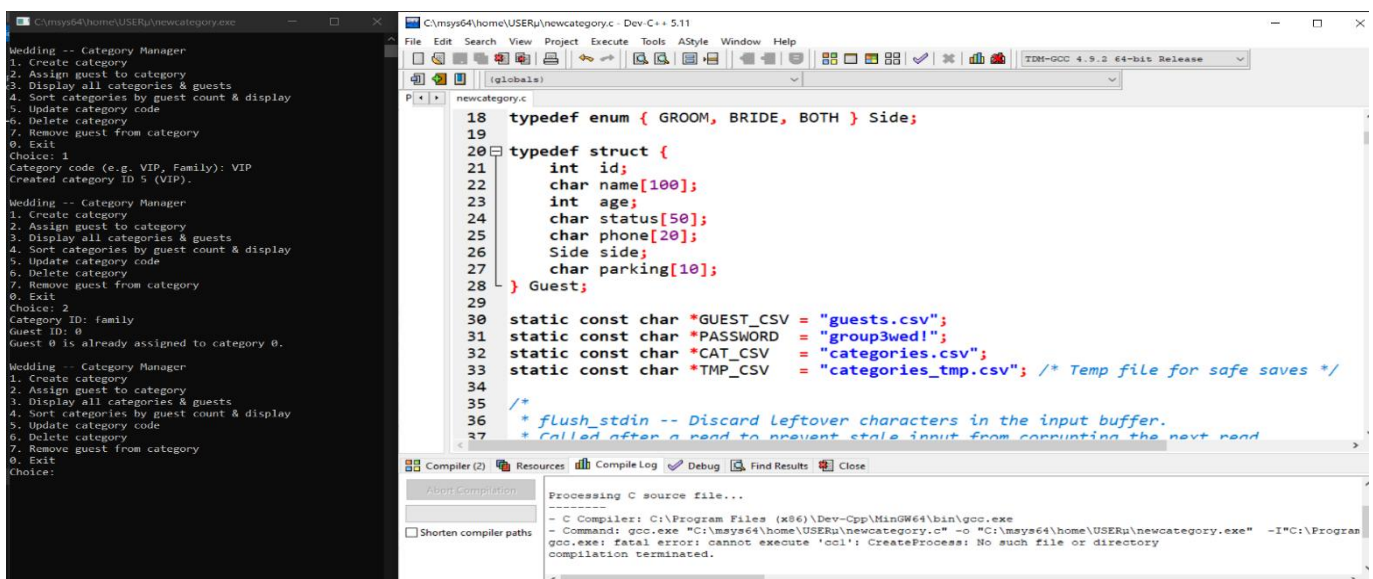
15.2 Category Management

The category management module was implemented in order to allow for creation of different categories at a wedding ceremony.

- Just like the person management module, alongside all the operations (Display, Update, Delete, Assign Guests)
- For assigning guests, the code was linked to the person management code, through the ID
- The code was thoroughly debugged using dev c++ software

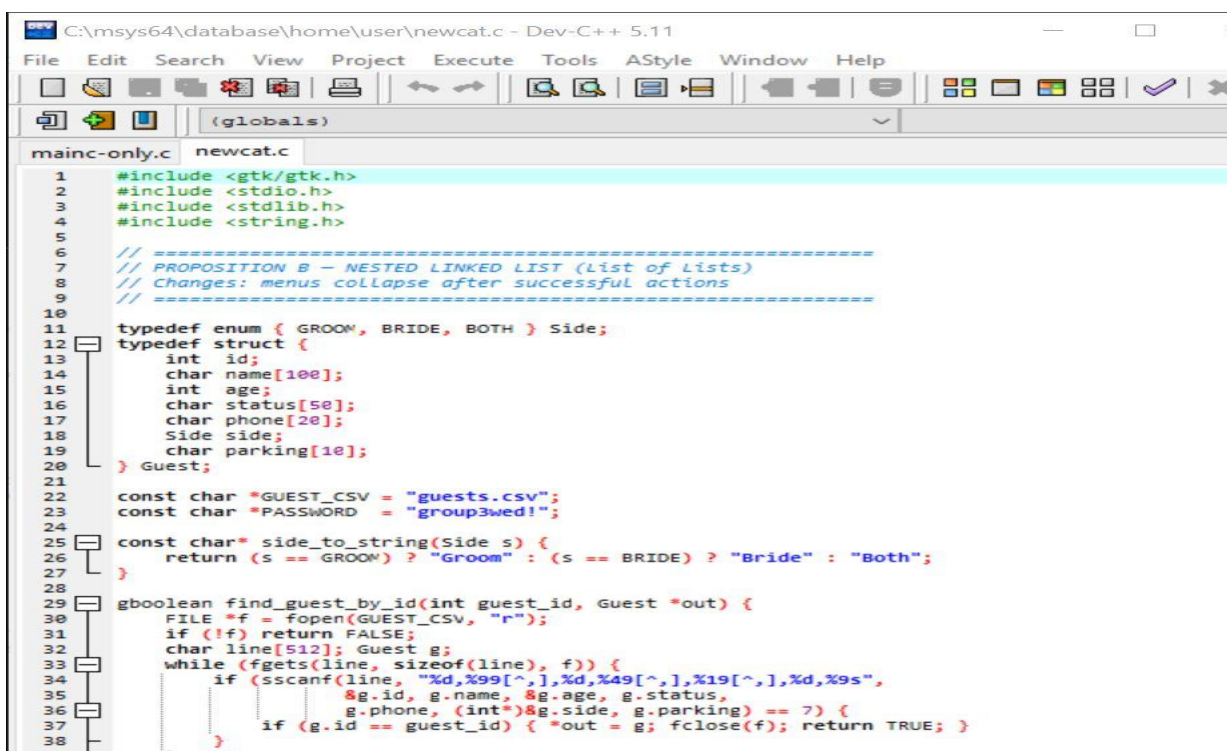
15.2.1 Extract of Category Management C-code

- Once the category management c program was correctly implemented, the GTK form version was created.



The screenshot shows the Dev-C++ IDE with two windows. The left window displays the 'Wedding -- Category Manager' menu, which includes options like '1. Create category', '2. Assign guest to category', '3. Display all categories & guests', '4. Sort categories by guest count & display', '5. Update category code', '6. Delete category', '7. Remove guest from category', and '0. Exit'. The right window shows the source code for 'newcategory.c'. The code defines a 'Side' enum with values GROOM, BRIDE, and BOTH, and a 'Guest' struct with fields for id, name, age, status, phone, side, and parking. It also defines several static constants for CSV files and a password. The bottom panel shows a compilation error: 'gcc.exe: fatal error: cannot execute 'cc1': CreateProcess: No such file or directory compilation terminated.'

15.2.2 Extract of Category Management GTK code



The screenshot shows the Dev-C++ IDE with the 'newcat.c' file open. The code includes headers for GTK, stdio, stdlib, and string. It defines a 'Side' enum and a 'Guest' struct. The code also includes a 'find_guest_by_id' function that searches for a guest by their ID in a CSV file. The code is written in C and uses GTK for the GUI.

en
perly
the
e, the

csv file was created automatically below the category management.

15.2.3 Category ran successfully

```
user@DESKTOP-S24HAFG UCRT64 ~
$ gcc newcat.c -o newcat.exe $(pkg-config --cflags --libs gtk4)

user@DESKTOP-S24HAFG UCRT64 ~
$ |
```

15.2.4 Executable file



 newcat	03/04/2026 09:42	C Source File	26 Ko
 newcat	03/04/2026 09:42	Application	168 Ko

- Haven properly created the csv file, test were carried out, similarly to those in the person management, to ensure functionability
- 15.2.4 Category Manager Window

15.2.5. Assignment Page

Wedding Category Manager [Nested List]

New Category

Assign Guest

Display

Update

Delete

? Close

Create a new Category

Code:

Create Category

Wedding Category Manager [Nested List]					
New Category	Assign Guest	Display	Update	Delete	
Password:					
Show All Sort by Guest Count					
NESTED LINKED LIST Categories: 1 Total guests assigned: 1 (No fixed guest limit per category)					
[Category 5 VIP] (1 guest(s))					
[GuestID 0] Matchim Celia Age: 18 VIP 67777777 Both Parking: Yes					

16. Combination of person and category management onto a single window

Haven made the person and category management fully-functional; it is necessary to combine them in a single window where a user has the possibility of choosing which to fill.

- The c-code, combinedmodules, was written, thoroughly de bugged and converted to a GTK form, and the csv file generated subsequently
- Main code with c only

16.1 Extract of combinedmodules.c

C:\Users\user\Desktop\pure person and category code\mainc-only.c - [Executing] - Dev...
File Edit Search View Project Execute Tools AStyle Window Help
(globals)
mainc-only.c

```

1 #include <stdio.h>
2 #include <stdlib.h>
3 #include <string.h>
4
5 /*
6  * Module descriptions
7  */
8 static void print_banner(void)
9 {
10     printf("\n");
11     printf("===== \n");
12     printf(" WEDDING GUEST SYSTEM - GROUP 3 \n");
13     printf(" Management Portal \n");
14     printf("===== \n");
15     printf("\n");
16 }
17
18 static void print_module_info(void)
19 {
20     printf(" MODULE 01 - Person Management \n");
21     printf(" Register, update and manage every wedding guest \n");
22     printf(" with full validation and CSV persistence. \n");
23     printf(" Features: \n");
24     printf("   - Add & save guests (CSV) \n");
25     printf("   - Live field validation \n");
26     printf("   - Update guest information \n");
27     printf("   - Password-protected delete \n");
28     printf("   - Auto-refresh display \n");
29     printf("\n");
30     printf(" MODULE 02 - Category Management \n");
31     printf(" Organise guests into nested linked-list \n");
32     printf(" categories such as VIP, Family, Friends. \n");
33     printf(" Features: \n");
34     printf("   - Create named categories \n");
35     printf("   - Assign guests (nested list) \n");
36     printf("   - Sort by guest count \n");
37     printf("   - Update & delete categories \n");
38     printf("   - Remove guests from category \n");
39     printf("\n");

```

```

MODULE 01 - Person Management
Register, update and manage every wedding guest
with full validation and CSV persistence.
Features:
  A Add & save guests (CSV)
  A Live field validation
  A Update guest information
  A Password-protected delete
  A Auto-refresh display

MODULE 02 - Category Management
Organise guests into nested linked-list
categories such as VIP, Family, Friends.
Features:
  A Create named categories
  A Assign guests (nested list)
  A Sort by guest count
  A Update & delete categories
  A Remove guests from category

PASSWORD PROTECTED - group3wed!
Wedding Guest System v1.0
-----

1. Open Module 01 - Person Management
2. Open Module 02 - Category Management
0. Exit

Choice:

```


16.2 Extract of Main code with Css

- Css was added to the GTK inorder to beautify the window

```
#include <gtk/gtk.h>
#include <stdlib.h>
#include <string.h>

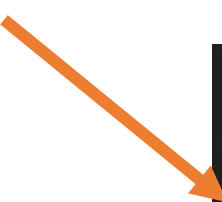
/*
   CSS - dark navy + gold palette
*/
static const char *APP_CSS =
/* window */
"window {"
"  background-color: #0d1b2e;"
"}"

/* Top banner */
".banner {"
"  background: linear-gradient(135deg, #091422 0%, #0d1b2e 60%, #112038 100%);"
"  padding: 36px 48px 28px 48px;"
"  border-bottom: 1px solid rgba(201,168,76,0.25);"
"}"
".banner-eyebrow {"
"  color: #c9a84c;"
"  font-size: 18px;"
"  letter-spacing: 4px;"
"  font-weight: bold;"
"}"
".banner-title {"
"  color: #f5f0e8;"
"  font-size: 30px;"
"  font-weight: 300;"
"  margin-top: 6px;"
"}"
".banner-sub {"
"  color: rgba(245,240,232,0.45);"
"  font-size: 12px;"
"  margin-top: 4px;"
"  letter-spacing: 1px;"
"}"
".gold-rule {"
"  background: linear-gradient(90deg, transparent, #c9a84c, transparent);"
"  min-height: 1px;"
"  margin: 16px 0 0 0;"
}
```

16.3 combinemodules ran Successfully

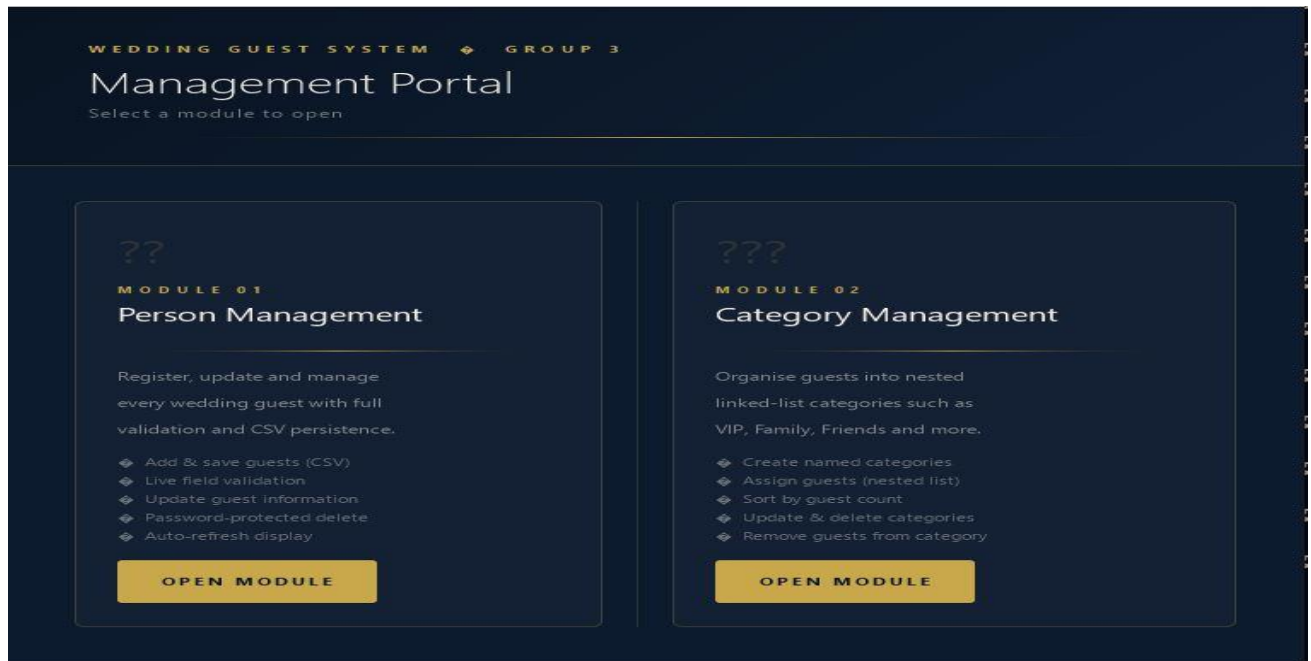
```
user@DESKTOP-S24HAFG UCRT64 ~
$ gcc mainapp.c -o mainapp.exe $(pkg-config --cflags --libs gtk4)
mainapp.c: In function 'on_launch_person':
mainapp.c:160:9: warning: 'gtk_message_dialog_new' is deprecated [-Wdeprecated-decl
arations]
  160 |         GtkWidget *dlg = gtk_message_dialog_new(
      |         ^~~~~~
In file included from C:/msys64/database/ucrt64/include/gtk-4.0/gtk/gtk.h:194,
      from mainapp.c:16:
C:/msys64/database/ucrt64/include/gtk-4.0/gtk/deprecated/gtkmessagedialog.h:81:12: n
ote: declared here
   81 | GtkWidget* gtk_message_dialog_new      (GtkWindow      *parent,
      |         ^~~~~~
mainapp.c: In function 'on_launch_category':
mainapp.c:181:9: warning: 'gtk_message_dialog_new' is deprecated [-Wdeprecated-decl
arations]
  181 |         GtkWidget *dlg = gtk_message_dialog_new(
      |         ^~~~~~
C:/msys64/database/ucrt64/include/gtk-4.0/gtk/deprecated/gtkmessagedialog.h:81:12: n
ote: declared here
   81 | GtkWidget* gtk_message_dialog_new      (GtkWindow      *parent,
      |         ^~~~~~
user@DESKTOP-S24HAFG UCRT64 ~
$ |
```

16.4. Executable file



 mainapp	03/04/2026 10:33	C Source File	13 Ko
 mainapp	03/04/2026 10:34	Application	153 Ko

16.5. Combinedmodules Window



17. THE GIFT MANAGEMENT

The gift management module was implemented in order manage the possible gifts at a wedding ceremony.

- It helps the couple to know what their guests are planning to give or offer them.
- Furthermore, just like the person and category management module, it carries alongsides all the operations;
 - Register
 - Display
 - Delete
 - update gifts from guests

17.1 Extract of Gift Management C-code

```
/* ACTION: Add / Register a gift */
/* ===== */
void action_add_gift(void) {
    printf("\n== Register a Gift ==\n");

    /* Step 1 = verify guest name (up to 3 tries) */
    char guest_name[100];
    int guest_id = -1;
    char category[50] = "Unknown";
    int verified = 0;

    for (int attempt = 0; attempt < 3 && !verified; attempt++) {
        read_line("Enter your full name (as registered): ", guest_name, sizeof(guest_name));
        if (verify_guest_name(guest_name, &guest_id, category, sizeof(category))) {
            verified = 1;
        } else {
            printf("Name not found in the guest list. Please re-enter the correct name.\n");
        }
    }

    if (!verified) {
        printf("Could not verify your name. Gift registration cancelled.\n");
        return;
    }

    printf("Welcome, %s! (ID: %d | Category: %s)\n", guest_name, guest_id, category);

    /* Step 2 = show catalogue and let guest choose */
}
```

```
***** Wedding Gift Management *****
1. Register a gift
2. Display all gifts
3. Delete a gift
4. Update a gift
0. Exit
=====
Choice: 2

=== Display All Gifts ===
Password: group3wed!
No gifts recorded yet.

***** Wedding Gift Management *****
1. Register a gift
2. Display all gifts
3. Delete a gift
4. Update a gift
0. Exit
=====
Choice:
```

```
***** Wedding Gift Management *****
=====
1. Register a gift
2. Display all gifts
3. Delete a gift
4. Update a gift
0. Exit
=====
Choice: 1

=== Register a Gift ===
Enter your full name (as registered): Brenda
Name not found in the guest list. Please re-enter the correct name.
Enter your full name (as registered):
```

- Once the gift management c program was correctly implemented, the GTK version was created

17.2 Extract of Gift Management GTK code

```
37 static const char *GUEST_CSV = "guests.csv";
38 static const char *CAT_CSV = "category.csv";
39 static const char *GIFT_CSV = "gifts.csv";
40 static const char *TMP_GIFT = "gifts_tmp.csv";
41 static const char *PASSWORD = "group3wed!";
42
43 typedef struct { int id; const char *name; double price_fcfa; } GiftItem;
44
45 static const GiftItem CATALOGUE[GIFT_ITEM_COUNT] = {
46     { 1, "House", 45000000.0 },
47     { 2, "Car", 8500000.0 },
48     { 3, "Pots Set", 35000.0 },
49     { 4, "Spoon Set", 8000.0 },
50     { 5, "Fork Set", 8000.0 },
51     { 6, "Dishes Set", 25000.0 },
52     { 7, "Glass Set", 15000.0 },
53     { 8, "Microwave", 55000.0 },
54     { 9, "Customized Photo Frame", 12000.0 },
55     { 10, "Travel Tickets (per person)", 350000.0 },
56     { 11, "Hotel Reservation (per night)", 120000.0 },
57     { 12, "Wine (bottle)", 18000.0 },
58     { 13, "Smoothing Iron", 22000.0 },
59     { 14, "Bedside Table", 45000.0 },
60     { 15, "Cushion Set", 20000.0 },
61     { 16, "Curtain Set", 30000.0 },
62     { 17, "Television (43\")", 180000.0 },
63     { 18, "Flower Pots", 7000.0 },
64     { 19, "Kettle", 15000.0 },
```

17.3 Gift management ran successfully

17.4 Executable file

```
USER@DESKTOP-KSHPOJE UCRT64 ~
$ gcc gift_gtk03.c -o gift_management_group3.exe $(pkg-config --cflags --libs gtk4) -lm

USER@DESKTOP-KSHPOJE UCRT64 ~
$ |
```

gift_management_group3.exe 04/03/2020 05:21 Application 192 Ko

- Haven properly created the csv file, test were carried out, similarly to those of category and person management so as to insure functionality.
- We then created the csv file in order to carryout the tests

csv file

gifts.csv 03/04/2026 06:39 XLS Worksheet 1 Ko

Register a gift

Wedding Gift Manager - group3wed

Register Gift Display Gifts Delete Gift Update Gift

No password required to register a gift. Enter your full name to verify your identity.

Step 1 : Verify your identity (Full Name)

Full Name:

Verified: Matchim Celia

Step 2 : Choose your gift

1. House 45000000 FCFA (68602.06 EUR) ▼

Unit price: 45000000 FCFA (68602.06 EUR)

Quantity: - +

Total: 45000000 FCFA (68602.06 EUR)

Step 3 : Thank-you name (optional)

- We implemented registration of a gift by a guest using their name because they do not know the id generated when they registered.
- In order to register a gift, a guest must have first registered in the person module.
- The name Matchim Celia, previously registered in the person.c was successfully identified, showing the success of our code

Delete Gift

Wedding Gift Manager - group3wed

Register Gift Display Gifts Delete Gift Update Gift

Step 1 : Password

Password accepted

Step 2 : Verify your identity (Full Name only)

Full Name:

Step 3 : Gift Record to delete

Record ID:

Display Gift

Password accepted

RID	Guest Name	Guest ID	Category	Gift Item
1	Matchim Celia	1	VIP	House

d

Gift Item	Qty	Total FCFA	Total EUR
House	1	45000000	68602,06

Update Gift

Wedding Gift Manager - group3wed

Register Gift Display Gifts Delete Gift Update Gift

Step 1 : Password

.....

Password accepted

Step 2 : Verify your identity (Full Name only)

Full Name:

Record ID:

Step 3 : Select new gift and quantity (total auto-calculated)

1. House 45000000 FCFA (68602.06 EUR) ▼

Unit price: 45000000 FCFA (68602.06 EUR)

Quantity: - +

Total: 45000000 FCFA (68602.06 EUR)

- To display, delete or update a gift, a password must be inserted without which these operations cannot be done.

NB:

- A password was used to prevent guests from accessing fields other than registration. Password: group3wed!

-For being lengthy, full codes could not be integrated in our report. Full codes are available in the GitHub repository at this link: <https://github.com/celia-pixel-rgb/Project-Wedding-Ceremony>

- All the outputs of the codes are present in the repository too.

Combinedmodules C Window

The screenshot displays a C++ IDE with two windows. The left window shows the source code for 'combinedmodules-1.c', and the right window shows the program's execution output.

Source Code (combinedmodules-1.c):

```
217
218 26 predefined gift options that guests can choose from.
219 The last option (index 25) is "Cash Contribution" with price 0,
220 because the guest enters a custom amount.
221
222 static char *giftNames[] = {
223     "House", "Car", "Pot Set", "Spoon Set", "Fork Set",
224     "Dish Set", "Glass Set", "Microwave", "Photo Frame",
225     "Travel Ticket", "Hotel Reservation", "Wine", "Iron",
226     "Bedside Table", "Cushion Set", "Curtain Set", "Television",
227     "Flower Pot", "Kettle", "Cup Set", "Sheet Set", "Jewelry",
228     "Couple Watch", "Home Decor", "Gas Stove", "Cash Contribution"
229 };
230
231 static float giftPrices[] = {
232     1500000, 800000, 50000, 10000, 10000,
233     30000, 20000, 70000, 15000,
234     200000, 150000, 10000,
235     25000, 40000, 30000, 35000,
236     250000, 10000, 15000, 12000,
237     30000, 100000, 80000, 50000,
238     60000, 0
239 };
240
241 /*
242 FUNCTION: gift_show_menu()
243
244 PURPOSE:
245 Displays the full list of 26 gift options with their prices.
246 Called before a guest makes or updates their gift choice.
247
248 PARAMETERS: void
249 RETURN VALUE: void
250 */
251 static void gift_show_menu(void)
252 {
253     int i;
254     printf("\n--- Gift List ---\n");
255     for (i = 0; i < 26; i++) {
```

Execution Output:

```
? Multiple payment methods
? Password-protected display
? Update & delete gift records

PASSWORD PROTECTED - group3wed!
Wedding Guest System v1.0
-----

1. Open Module 01 - Person Management
2. Open Module 02 - Category Management
3. Open Module 03 - Gift Management
0. Exit

Choice: 3

=====
MODULE 03 - Gift Management
=====

1. Add a gift
2. Display all gifts
3. Display one gift
4. Update a gift
5. Delete a gift
0. Return to Main Menu

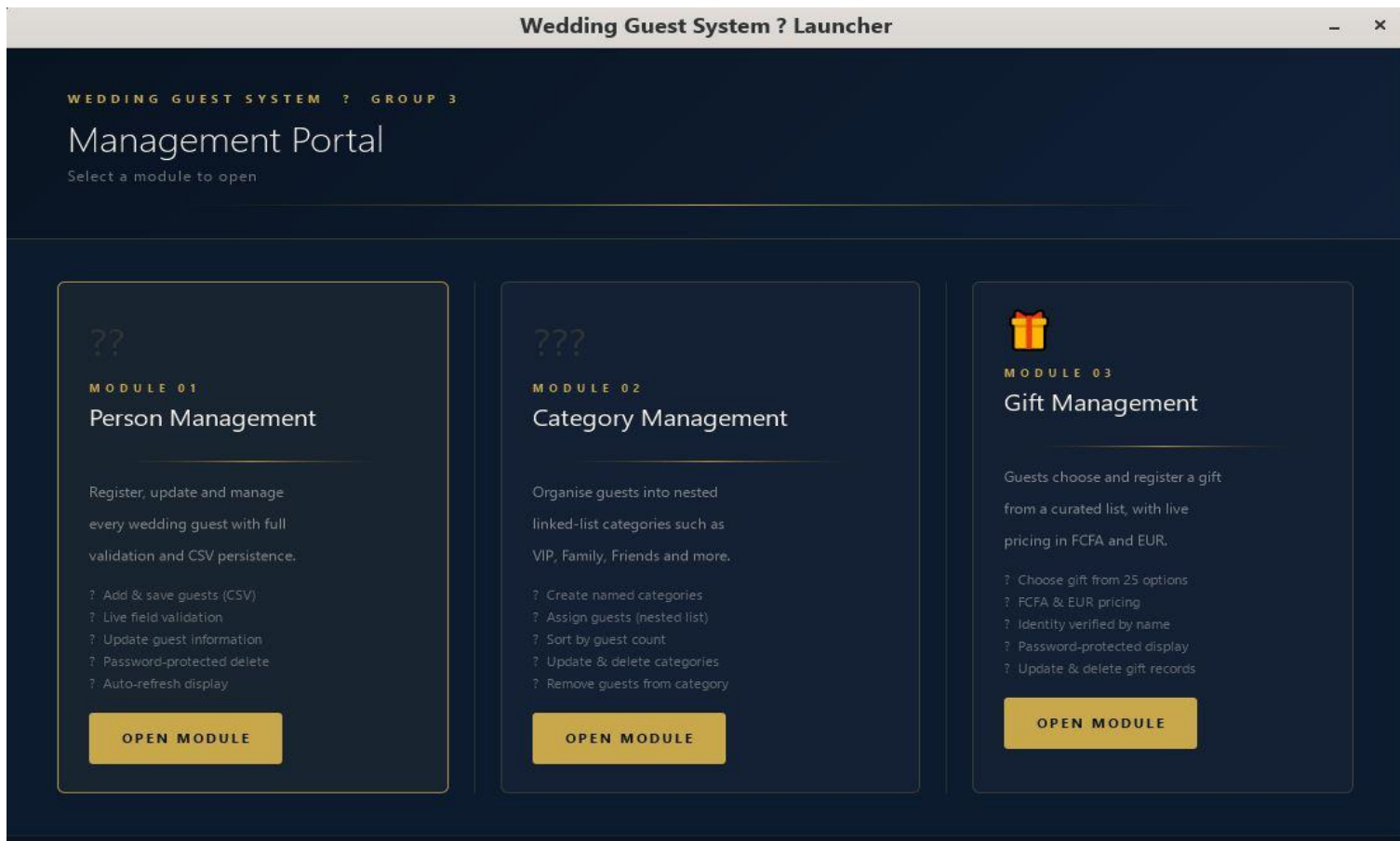
Choice: 2
Enter password: group3wed!
No gift records found.

=====
MODULE 03 - Gift Management
=====

1. Add a gift
2. Display all gifts
3. Display one gift
4. Update a gift
5. Delete a gift
0. Return to Main Menu

Choice:
```

✓ We then added the gift management to our combinedmodules.c

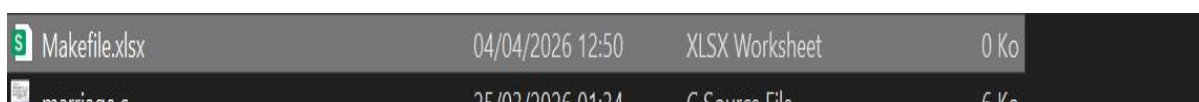


Combinedmodules GTK Window

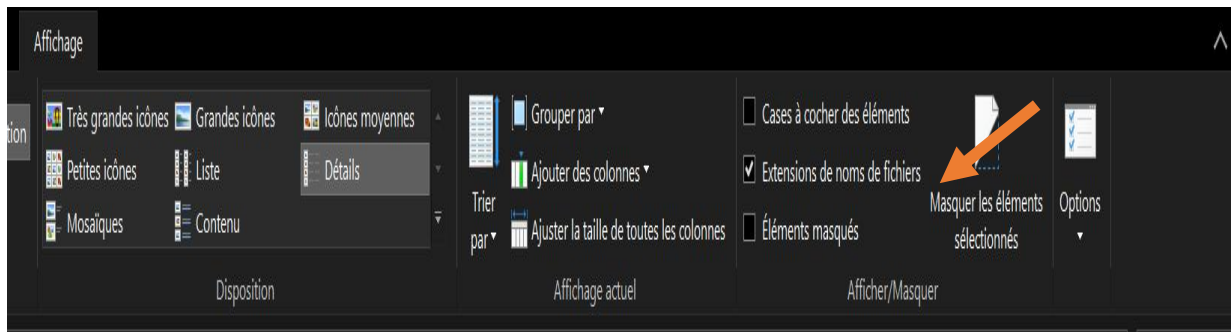
18. CREATION OF MAKE FILE

A Makefile is a special text file used with a build automation tool called Make. It tells the tool how to compile, build, and manage a project.

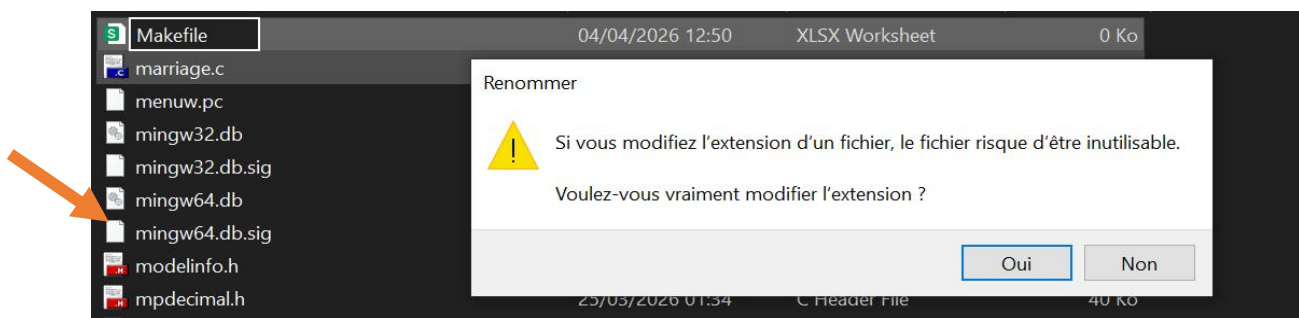
- To better manage our program, we will build a Makefile in order to corelate all the modules.
- Create a file you name Makefile in the same folder as your c programs. You can create it as word, excel, blocnotes etc



- Remove the extension of the file by going to the “Affichage” component and check the box “show ext and safeguard changes, so that your make file has no extension.



- Modify the extension by clicking “Oui” or “Yes.



- Your Makefile has been created, you can now proceed with writing information into it.
- Open the MSYS2UCRT64 terminal and type in the command to search for the path where your Makefile is found

```
user@DESKTOP-S24HAFG UCRT64 ~
$ cd C:/msys64/database/home/user

user@DESKTOP-S24HAFG UCRT64 /c/msys64/database/home/user
$ |
```

- Type “make” in the terminal, and wait until all the different modules are all compiled.

```
[OK] launcher compiled

=====
All targets built successfully!
=====

user@DESKTOP-S24HAFG UCRT64 ~
$ |
```

- After your Makefile ran correctly, you can now proceed to launch the interface created with all your modules by typing the command “./launcher” in your terminal

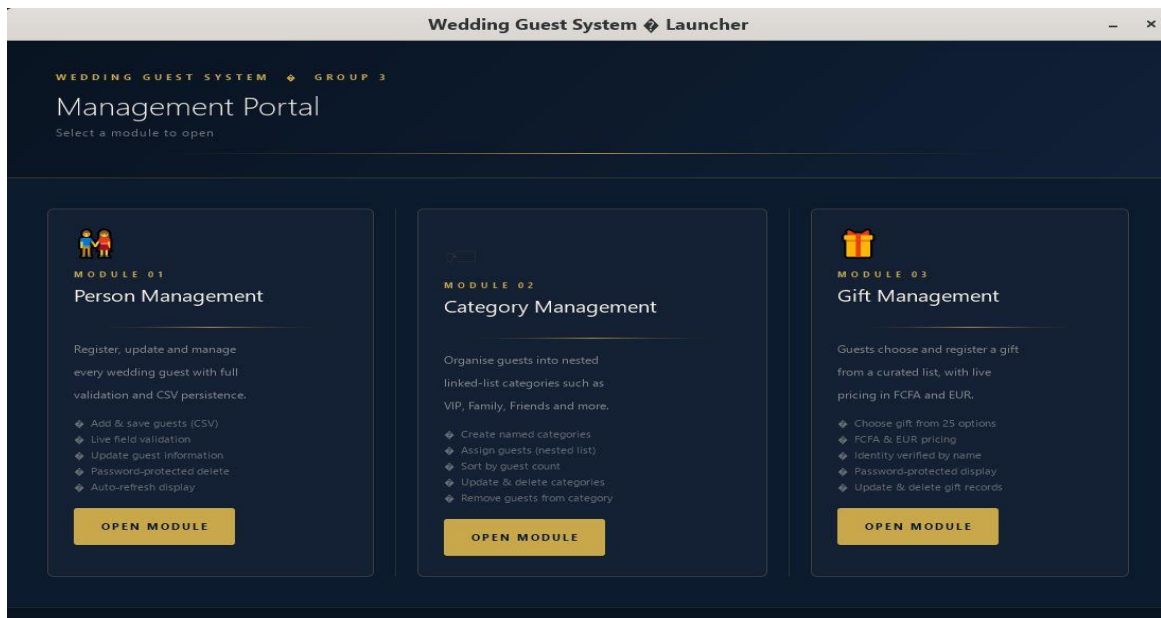
```
user@DESKTOP-S24HAFG UCRT64 ~
$ ./launcher

(launcher.exe:5512): GLib-WARNING **: 08:32:21.092: Failed to determine console output code page: Le système n'a pas trouvé l'option d'environnement spécifiée.. Falling back to UTF-8

(launcher.exe:5512): Pango-WARNING **: 08:32:21.130: Invalid UTF-8 string passed to pango_layout_set_text()

(launcher.exe:5512): Pango-WARNING **: 08:32:21.149: Invalid UTF-8 string passed to pango_layout_set_text()
```

- Once it is done, a cube shaped GTK icon appears in your task bar. Click on it, you will see the interface with all three modules.



- You use the make clean prompt in order to clear away any actions done in the modules

```
User@DESKTOP-S24HAFG UCRT64 ~
$ make clean
rm -f person category gift combinedmodules \
  gtkperson gtkcategory gift_gtk03 launcher \
  temp.csv temp_renum.csv category_tmp.csv gifts_tmp.csv
[OK] clean done
```

Optimizations Made

- We rendered the id of a person dynamic that is, if someone registers as a guest and has ID 0, and later on with draws, the ID of the persons following will immediately be adjusted; i.e the next will go to 0, and the following to 1...

❖ Github link: <https://github.com/celia-pixel-rgb/Project-Wedding-Ceremony>

19. CONCLUSION

With Modules 1,2 and 3 implemented and optimized, we are ready for the implementation of module 4 and 5, parking management and priority management.